

Ecliptic Watchdogs of Intelligent Perception Smart IoT Surveillance for Women Protection

A Project Report

Submitted in partial fulfillment of the Requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

ELECTRONICS AND COMMUNICATION ENGINEERING

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES
(Autonomous)**

(Approved by AICTE New Delhi || Permanently Affiliated to
JNTUK, Kakinada) || Accredited with 'A' Grade by NAAC ||
NBA Accreditation)

Vinjanampadu (V), Vatticherukuru (M), Guntur (Dt), A.P-522017.

APRIL – 2026.

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CERTIFICATE

This is to certify that this project report entitled **“Ecliptic Watchdogs of Intelligent Perception Smart IoT Surveillance for Women Protection”** submitted by REBBA VENKATESH (22JR1A04F2), YEDIDA SAI VISWANADH (22JR1A04H2), YENDLURI HEMANTH (22JR1A04H4), PUPPALA VENKATA KISHORE (22JR1A04F0), SHAIK MOHAMMAD SAJID (22JR1A04F8) to Jawaharlal Nehru University Kakinada, through KKR & KSR Institute of Technology and Sciences for the award of the Degree of Bachelor of Technology in **ELECTRONICS AND COMMUNICATION ENGINEERING** is a Bonafide record of project work carried out under the supervision of Dr M PURNA KISHORE during the year 2025-2026.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the project **“Ecliptic Watchdogs of Intelligent Perception Smart IoT Surveillance for Women Protection”** has been carried out by me and this work has been submitted to KKR & KSR Institute of Technology and Sciences (A), Vinjanampadu, affiliated to Jawaharlal Nehru Technological University, Kakinada in partial fulfillment of the requirements for the award of degree of Bachelor of Technology in **Electronics and Communication Engineering**.

We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

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- To support the cross-cultural exchange of knowledge between industry and academy.
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- Inculcate professional behavior, and strong ethical values to meet the challenges in collaboration and lifelong learning.

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Apply Electronics, Communications, VLSI, Embedded systems knowledge to arrive cost effective and appropriate solutions.

PSO2: Problem-solving skills:

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PSO3: Successful career:

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PSO4: Exposure for research and development:

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PEO4:

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PEO5:

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Broad application of mathematics, science, engineering fundamentals, and specialization to solve complex problems.

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PO4: Conduct Investigations of Complex Problems

Application of research methods—including experimentation, data analysis, interpretation, and synthesis—to derive valid conclusions.

PO5: Engineering Tool Usage

Creation, selection, and application of appropriate techniques and modern engineering/IT tools (e.g., modeling, prediction), with awareness of their limitations.

PO6: The Engineer and The Work

Merges the roles of engineer in society with environmental responsibility—a fusion of the previous “Engineer & Society” and “Environment & Sustainability” POs..

PO7: Ethics

Ethical reasoning in decision-making, upholding professional ethics and inclusive behavior (formerly PO8, but refocused to include inclusivity).

PO8: Individual and Collaborative Teamwork

Effective functioning as an individual or as part of diverse, multidisciplinary teams (adjusts wording from previous PO9).

PO9: Communication

Effective communication—oral, written, and graphical—with engineering community and society at large—with a focus on inclusive and clear messaging.

PO10: Project Management and Finance

Understanding and application of engineering management principles to one's own work, suitable for team leadership and multidisciplinary environments.

PO11: Life-Long Learning

Recognition of, and readiness for, continuous and adaptive learning in an evolving technological landscape

COURSE OUTCOMES (CO'S)

CO409.1: Perform a system of examinations to identify problems.

CO409.2: Review the literature/Related work.

CO409.3: Defining the problem & its area of domain.

CO409.4: Proposal of solution for the selected area/methodology.

CO409.5: Analysis of the proposed work & documentation.

CO409.6: Acquire collaborative learning, leadership qualities & presentation skills

COURSE OUTCOMES - PROGRAM OUTCOMES MAPPING

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO409.1	2	3		2		2	2				
CO409.2	3	2	3	3		3			2	2	3
CO409.3	3	3	3	3	2	2					
CO409.4	3	3	2	2	3		2	2	1	3	
CO409.5	3		3	2	2		2		2		1
CO409.6	1					3				2	
Average	3	3	3	3	3	3	2	2	2	3	2

3: High 2: Medium 1: Low

COURSE OUTCOMES - PROGRAM SPECIFIC OUTCOMES MAPPING

CO code	PSO1	PSO2	PSO3	PSO4
CO409.1	3			
CO409.2	3	2	1	
CO409.3	3		2	2
CO409.4	3		2	
CO409.5	3	2		
CO409.6		3		
Average	3	2	2	2

High 2: Medium 1: Low

PROGRAM EDUCATIONAL OBJECTIVES – PROGRAM SPECIFIC OUTCOMES MAPPING

	PSO1	PSO2	PSO3	PSO4
PEO1	2	1	3	1
PEO2		3	2	2
PEO3	1	2	3	3
PEO4	3		2	2
PEO5	1	3	2	
Average	1	2	2	2

3: High 2: Medium 1: Low

CO-PO'S MAPPING WITH REASONS

1. **CO409.1** is mapped with PO1, PO2 and PO4, PO6, PO7 as basic knowledge of Engineering and problem Analysis activities are highly essential to conduct examinations on existing systems which have been using in industries as a part of and to define the problem of the proposed system.
2. **CO409.2** is mapped with PO1, PO2, PO3, PO4 and PO6, PO9, PO10, and PO11 for identification, gathering analysis, and classification of requirements for the proposed system, basic knowledge of engineering and Analysis steps along with complex problem analysis through the efforts of teamwork in order to meet the specific needs of the customer.
3. **CO409.3** is mapped with PO1, PO2, PO3, PO4, PO5, PO6 and PO12 as to conduct the literature review and to examine the relevant systems to understand and identify the merits and demerits of each to enhance and develop the proposed as per the need.
4. **CO409.4** is mapped with PO1, PO2, PO3, PO4, PO5, PO7, PO8, PO9, and PO10 because modularization and design of the project are needed after requirements elicitation. For modularization and design of the project, Basic knowledge of Engineering, Analysis capabilities, Design skills, and communication is needed between team members as different modules are designed individually before integration.
5. **CO409.5** is mapped with PO1, PO3, PO4, PO5, PO7, PO9, PO11 and PO12 as to construct the project latest technologies are needed. The development of the project is done individually and in groups with well-defined communication by using engineering and management principles.
4. **CO409.6** is mapped with PO1, PO6, PO10, and PO12 because during and after completion of the project, documentation is needed along with proper methods of

presentation through understanding and application of engineering and management principles, which in turn needs well-defined communication between the team members with all the ethical values. Even the project development team defines future enhancements as a part of the project development after identifying the scope of the project.

CO-PSO'S MAPPING WITH REASONS

1. **CO409.1** is mapped with **PSO1** as examining existing systems and identification of the problem is a part of Application Development activity and identification of evolutionary changes in the latest technologies.
2. **CO409.2** is mapped with **PSO1, PSO2, and PSO3** as identifying and classifying the requirements is a part of Application development and evolutionary computing changes and also follows ethical principles.
3. **CO409.3** is mapped with **PSO1 and PSO3, PSO4** as a review of literature is a part of application development activity by recognizing the computing technologies and their evolutionary changes.
4. **CO409.4** is mapped with **PSO1 and PSO3** because modularization and logical design is also a part of Application development and follows computing changes using Deep learning technology.
5. **CO409.5** is mapped with **PSO1 and PSO2** as Testing, Development, and Integration of project activities are part of Application development and follow ethical principles.
6. **CO409.6** is mapped with **PSO2** for project documentation and presentation the project team members apply the professional and leadership quality

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ABSTRACT

The IoT-Powered Night Patrolling System for Women Safety is an advanced security solution developed to enhance safety and surveillance in vulnerable and low-visibility areas, especially during nighttime. In many regions, limited patrolling, poor lighting conditions, and delayed response systems create serious risks for women.

The system is built around a Raspberry Pi, which acts as the central processing unit. A high-definition webcam is connected to the Raspberry Pi to continuously capture live video streams of the monitored environment. The captured video frames are processed using a Python-based application that implements face detection and recognition algorithms. Typically, techniques such as Haar Cascade classifiers for face detection and Local Binary Patterns Histogram (LBPH) or deep learning models are used for accurate identification of individuals.

In such cases, the system immediately performs a series of automated actions. It captures a clear image of the detected individual and generates an alert notification. Using IoT connectivity (via Wi-Fi or mobile network), the captured image along with relevant information such as timestamp and location is sent to a pre-registered email address or cloud platform. This ensures that concerned authorities, security personnel, or guardians receive real-time updates and can take prompt action if required.

Additionally, the system can be extended with features such as motion detection, cloud storage, mobile app integration, and GPS tracking for enhanced functionality. These improvements allow for better monitoring, remote access, and historical data analysis, making the system more robust and scalable. The use of IoT also enables multiple devices to be interconnected, forming a network of smart surveillance units across larger areas.

Overall, the IoT-Powered Night Patrolling System serves as an intelligent surveillance solution that leverages modern technologies to enhance women's safety. It not only detects suspicious activities but also ensures timely communication and response, thereby contributing to a safer and more secure environment.

Keywords: Women Safety, Internet of Things (IoT), Computer Vision, Face Recognition, Raspberry Pi, Smart Surveillance, Real-Time Monitoring.

CHAPTER-1

Introduction

1.1 Introduction

Women's safety has become a major concern in many parts of the world, particularly during nighttime when public areas may have limited security and reduced human presence. Incidents of harassment, assault, and other crimes often occur in poorly monitored environments such as streets, parking areas, campuses, and isolated locations.

The **IoT-Powered Night Patrolling System for Women Safety** is proposed as a smart surveillance solution designed to monitor sensitive areas and detect unauthorized individuals. The system integrates computer vision and Internet of Things (IoT) technologies to automatically observe and analyse activities in a monitored environment. A Python-based program runs on the Raspberry Pi, which acts as the central processing unit for capturing and analysing video frames from a connected webcam.

The system uses face recognition techniques to identify individuals present in the monitored area. A database containing images of authorized individuals is created and stored in the system. When a person enters the monitored region, the camera captures the face and compares it with the stored database. If the face matches an authorized individual, the system continues monitoring without triggering any alert. However, if an unknown or unauthorized person is detected, the system treats it as a potential security concern.

In such situations, the system automatically captures the image of the unidentified individual and sends an alert notification. The captured image is transmitted to a pre-registered email address so that security personnel, authorities, or guardians can take immediate action if necessary. This real-time alert mechanism enables faster response and helps prevent potential threats before they escalate.

By combining IoT connectivity with computer vision-based facial recognition, the proposed system provides a cost-effective and intelligent security solution.

1.2 Embedded System Implementation

An embedded system is one kind of a computer system mainly designed to perform several tasks like to access, process, and store and also control the data in various electronics-based systems. Embedded systems are a combination of hardware and software where software is usually known as firmware that is embedded into the hardware. One of its most important characteristics of these systems is, it gives the o/p within the time limits. Embedded systems support to make the work more perfect and convenient. So, we frequently use embedded systems in simple and complex devices too. The applications of embedded systems mainly involve in our real life for several devices like microwave, calculators, TV remote control, home security and neighborhood traffic control systems, etc.

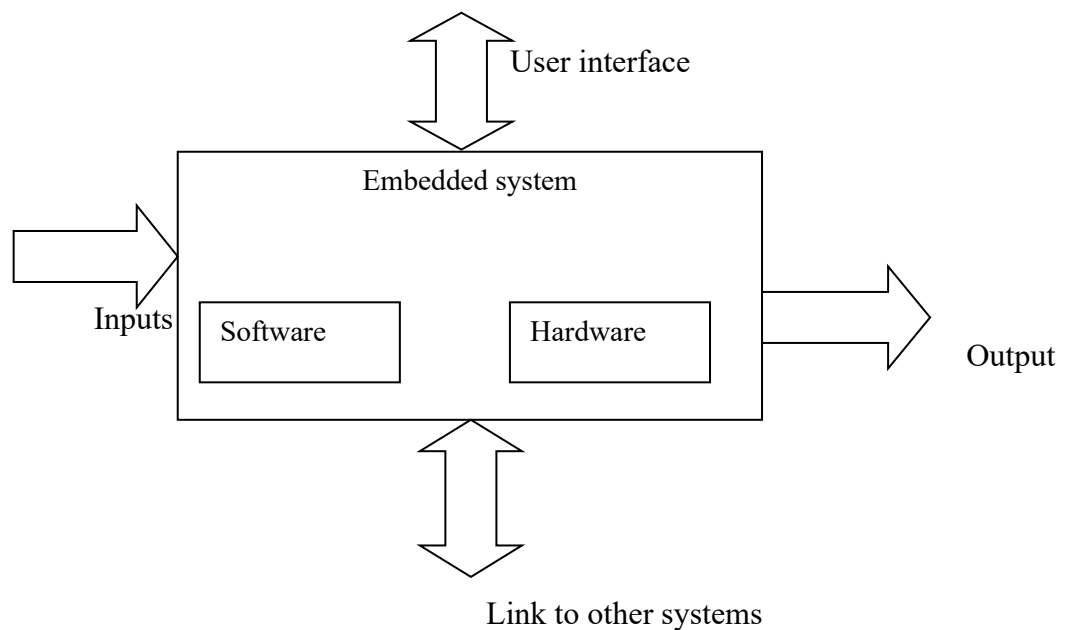


Fig 1: Overview of embedded systems

1.3 Embedded Systems:

Embedded system includes mainly two sections, they are

1. Hardware
2. Software

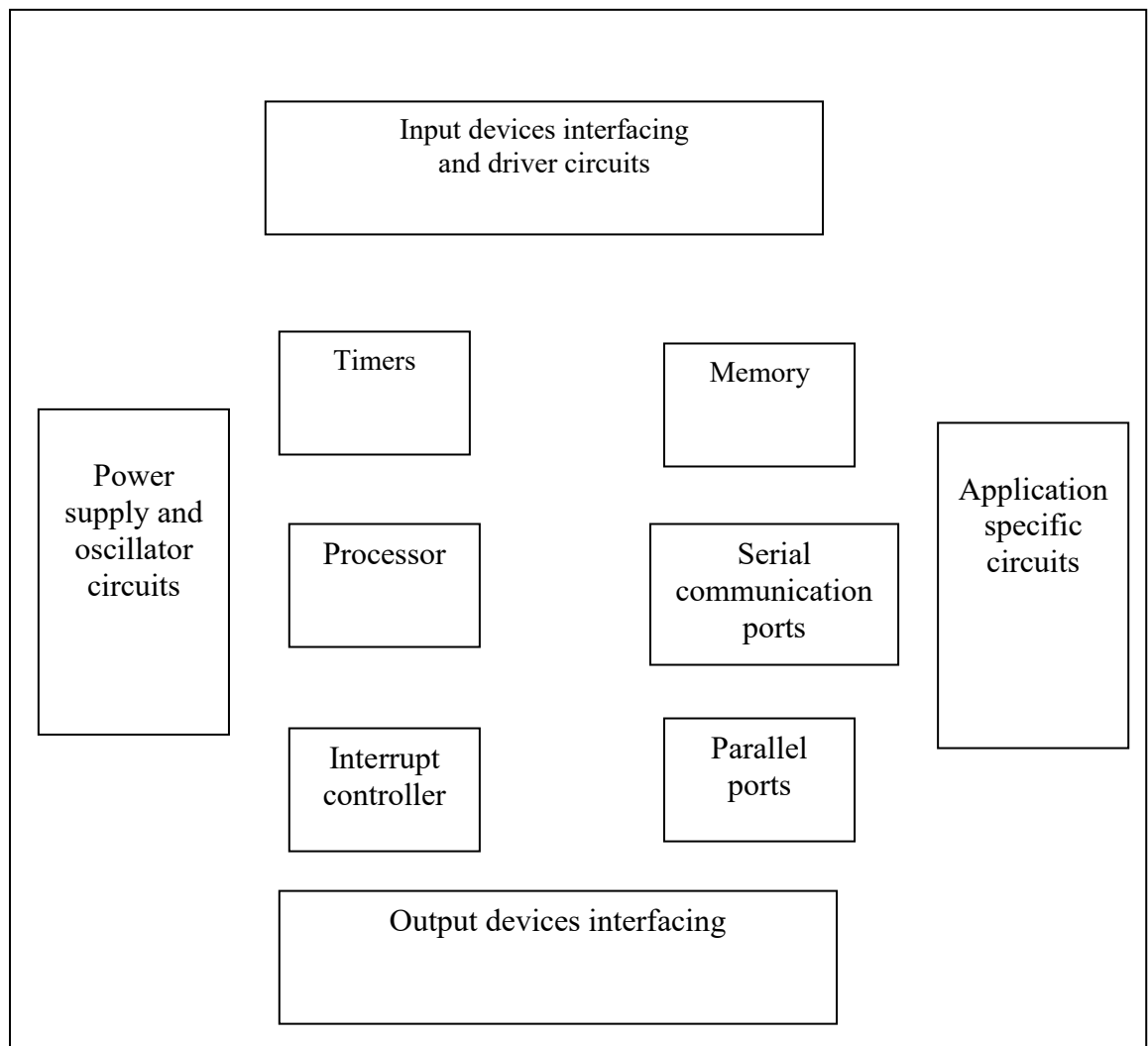


Fig 2: Internal Overview of embedded system

1.4 Embedded System Hardware:

As with any electronic system, an embedded system requires a hardware platform on which it performs the operation. Embedded system hardware is built with a microprocessor or microcontroller. The embedded system hardware has elements like input output (I/O) interfaces, user interface, memory and the display. Usually, an embedded system consists of:

Power Supply

- Processor
- Memory
- Timers
- Serial communication ports
- Output/Output circuits
- System application specific circuits

Embedded systems use different processors for its desired operation. Some of the processors used are

1. Microprocessor
2. Microcontroller
3. Digital signal processor

Microprocessor vs. Microcontroller

Microprocessor:

- CPU on a chip.
- We can attach required amount of ROM, RAM and I/O ports.
- Expensive due to external peripherals.
- Large in size
- general-purpose

Microcontroller :

- Computer on a chip
- fixed amount of on-chip ROM, RAM, I/O ports
- Low cost.
- Compact in size.
- Specific –purpose

Embedded System Software:

The embedded system software is written to perform a specific function. It is typically written in a high level format and then compiled down to provide code that can be lodged within a non-volatile memory within the hardware. An embedded system software is designed to keep in view of the three limits:

- Availability of system memory
- Availability of processor's speed
- When the system runs continuously, there is a need to limit power dissipation for events like stop, run and wake up.

Bringing software and hardware together for embedded system:

To make software to work with embedded systems we need to bring software and hardware together for this purpose we need to burn our source code into microprocessor or microcontroller which is a hardware component and which takes care of all operations to be done by embedded system according to our code.

Generally, we write source codes for embedded systems in assembly language, but the processors run only executable files. The process of converting the source code representation of your embedded software into an executable binary image involves three distinct steps:

- 1) Each of the source files must be compiled or assembled into an object file.
- 2) All of the object files that result from the first step must be linked together to produce a single object file, called the re-locatable program.
- 3) Physical memory addresses must be assigned to the relative offsets within the re-locatable program in a process called relocation.
- 4) The result of the final step is a file containing an executable binary image that is ready to run on the embedded system.

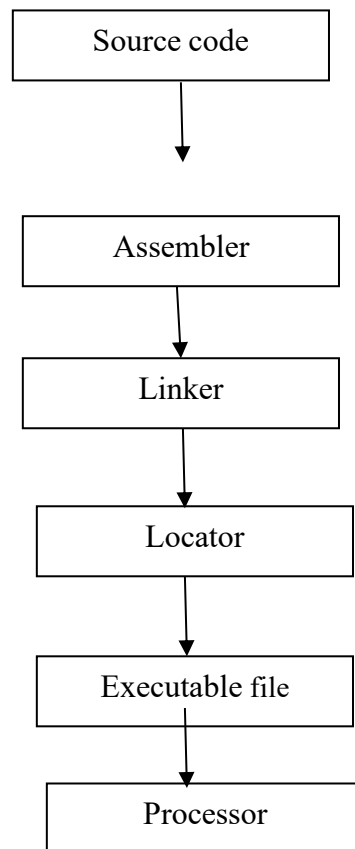


Fig 3: Flow of burning source code to processor

The process of preparing an embedded system program involves several stages that transform human-readable source code into a machine-executable binary image. Each stage plays a crucial role in ensuring that the program runs correctly on the target hardware.

1.5 Applications:

Embedded systems have different applications. A few select applications of embedded systems are smart cards, telecommunications, satellites, missiles, digital consumer electronics, computer networking, etc.

Embedded Systems in Automobiles

- Motor Control System
- Engine or Body Safety
- Robotics in Assembly Line
- Mobile and E-Com Access

Embedded systems in Telecommunications

- Mobile computing
- Networking
- Wireless Communications

Embedded Systems in Smart Cards

- Banking
- Telephone
- Security Systems

Implementation flow:

Stage 1:

Considering the problems of existing methods and giving solution to that problem by considering the basic requirements for our proposed system

Stage 2:

Considering the hardware requirement for the proposed system

For this we need to select the below components:

1. Microcontroller
2. Inputs for the proposed system (ex: sensors, drivers etc...)
3. Outputs (ex: relays, loads)

Stage 3:

After considering hardware requirements, now we need to check out the software requirements. Based on the microcontroller we select there exists different software for coding, compiling, debugging. we need to write source code for that proposed system based on our requirements and compile, debug the code in that software.

After completing all the requirements of software and hardware we need to bring both together to work our system. For this we need to burn our source code into microcontroller, after burning our source code to microcontroller then connect all input and output modules as per our requirement.

1.6 EXISTING SYSTEM:

In traditional security systems for women's safety, monitoring is mainly carried out through manual night patrolling and the use of closed-circuit television (CCTV) cameras. Security personnel are assigned to patrol specific areas during nighttime to observe suspicious activities and ensure public safety. Although CCTV cameras help in recording activities in different locations, they generally require continuous human monitoring. Security operators must watch the live video feeds to detect any unusual behavior, which can be time-consuming and prone to human error, especially when monitoring multiple cameras simultaneously.

Another common approach used in safety systems is the installation of alarm-based security devices and emergency alert mechanisms. In many places, women are provided with mobile applications or panic buttons that allow them to send distress signals when they feel unsafe. While these solutions help in emergency situations, they depend heavily on the individual's ability to activate the alert system during a threat. In many cases, the victim may not have enough time or opportunity to trigger the alarm, reducing the effectiveness of such systems.

1.7 DRAWBACKS:

1. Dependence on Manual Monitoring

Traditional CCTV surveillance systems require security personnel to continuously monitor video feeds. This manual monitoring can lead to fatigue and reduced attention over long periods, increasing the chances of missing suspicious activities.

2. Delay in Threat Detection

In many traditional systems, threats are identified only after an incident has occurred. Security teams often review recorded footage later, which prevents immediate action and reduces the effectiveness of the surveillance system.

3. Limited Night Patrolling Coverage

Human night patrolling cannot cover all areas continuously, especially in large campuses, streets, or isolated locations. Some areas may remain unmonitored for long periods, creating opportunities for criminal activities.

4. Human Errors in Surveillance

Security personnel may sometimes overlook suspicious individuals due to distractions, fatigue, or lack of concentration. This human limitation can reduce the reliability of traditional monitoring systems.

5. Lack of Intelligent Identification

Conventional surveillance systems record video but do not automatically recognize or identify individuals. Without technologies like face recognition, the system cannot distinguish between authorized persons and unknown individuals.

6. Slow Response to Emergencies

Many existing safety systems rely on manual reporting or emergency calls. This can delay the response time, especially if the incident occurs in a remote area where immediate communication is difficult.

1.8 PROPOSED SYSTEM

The proposed IoT-Powered Night Patrolling System for Women Safety aims to provide an intelligent and automated surveillance solution using modern technologies such as Internet of Things (IoT) and computer vision. The system is designed to continuously monitor a specific area during nighttime and detect the presence of individuals entering the monitored zone. A Python-based program runs on the Raspberry Pi, which acts as the main processing unit. A webcam connected to the system captures live video frames, enabling continuous observation of the surrounding environment.

In the proposed system, face recognition technology is used to identify individuals. A database containing images of authorized persons is created and stored in the system. When the camera captures a face, the system processes the image and compares it with the stored database using computer vision algorithms. If the detected face matches one of the authorized faces in the database, the system continues monitoring without any action. However, if the face is not recognized or does not match the stored database, the system identifies the individual as an unknown person.

Once an unknown person is detected, the system immediately captures the image of that individual and generates an alert. The captured image is automatically sent to a pre-registered email address using an automated notification system. This feature ensures that responsible authorities, security personnel, or guardians are informed instantly about the presence of a suspicious person in the monitored area.

By integrating IoT connectivity with real-time facial recognition, the proposed system provides a smart, efficient, and low-cost surveillance mechanism. The system can be installed in locations such as campuses, hostels, streets, and public areas where nighttime security is important. The automated monitoring and instant alert mechanism help in preventing potential threats and improving safety measures for women by enabling quick response from authorities.

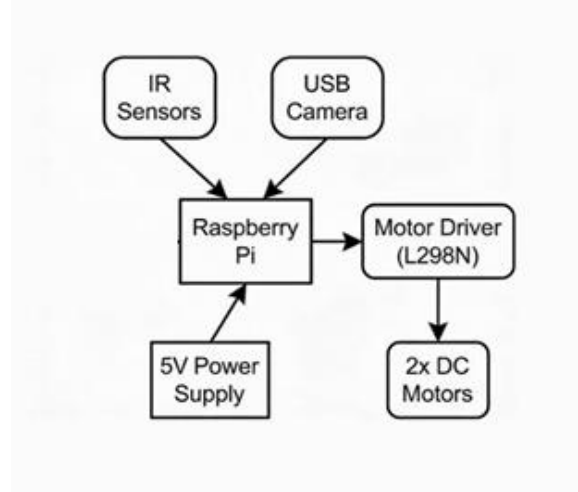


Fig 4: Block Diagram of the Proposed System

1.9 ADVANTAGES

1. Real-Time Monitoring

The proposed system continuously monitors the surrounding area using a camera connected to the Raspberry Pi. Unlike traditional systems that require manual observation, this system automatically analyzes video frames in real time. Continuous monitoring ensures that suspicious activities or unauthorized individuals are detected immediately, improving the overall security of the area.

2. Automatic Face Recognition

The system uses computer vision algorithms to identify individuals through facial recognition. By comparing detected faces with a stored database of authorized persons, the system can automatically differentiate between known and unknown individuals. This intelligent identification helps in detecting suspicious persons quickly without human intervention.

3. Instant Alert Generation

When an unknown person is detected, the system immediately captures the image and sends it to a pre-registered email address. This instant alert mechanism allows authorities, guardians, or security personnel to take quick action. Early alerts can help prevent potential crimes or dangerous situations.

4. Reduced Human Effort

Traditional surveillance systems require security personnel to constantly watch CCTV footage. The proposed system automates the monitoring and detection process. This significantly reduces the workload of security staff and allows them to focus on responding to alerts instead of continuously observing cameras.

5. Improved Women Safety

The primary objective of the system is to enhance safety for women in isolated or poorly monitored areas, especially during nighttime. By automatically detecting unfamiliar individuals and sending alerts, the system provides an additional layer of protection and helps create a safer environment.

6. Cost-Effective Solution

The system is built using affordable hardware components such as a webcam and a Raspberry Pi. Compared to complex security infrastructure, this setup is relatively low cost and easy to deploy. Institutions, campuses, and public areas can implement this system without requiring expensive equipment.

7. Automated Evidence Collection

When an unknown person is detected, the system captures and stores the image of the individual. This captured image can serve as useful evidence in case of suspicious activities or criminal investigations. It helps authorities identify suspects quickly.

8. Continuous Night Surveillance

Human patrol teams cannot monitor all areas continuously during nighttime. However, the proposed system operates 24/7 without fatigue or interruptions. This ensures that sensitive areas remain under constant surveillance even when human monitoring is limited.

9. Easy Integration with IoT Technologies

The system can be easily integrated with other IoT technologies such as cloud storage, mobile alerts, or additional sensors. This flexibility allows the system to be expanded in the future to include more advanced safety features.

10. Quick Response and Prevention

Since the system automatically detects unknown individuals and sends alerts immediately, it enables faster response from authorities. Quick response helps prevent potential threats before they escalate into serious incidents, making the environment safer for women.

1.10 REAL WORLD IMPLEMENTATIONS

1. **Women Safety in Public Streets** One of the most important applications of this system is improving safety for women in public streets, especially during nighttime. Many streets and residential areas have limited police patrol or security presence after dark. By installing this system in such locations, continuous monitoring can be achieved through automated surveillance. The camera connected to the Raspberry Pi captures live video and performs face recognition. If an unknown or suspicious individual enters the monitored area, the system immediately captures the image and sends an alert to authorities. This proactive monitoring helps reduce the chances of harassment or crimes against women in public places.

2. **University and College Campuses** Educational institutions such as universities and colleges often have large campuses where students move around during late hours for study or hostel activities. Ensuring the safety of female students in these environments is very important. The proposed system can be installed at important locations like hostels, libraries, corridors, and campus roads. It helps identify unknown individuals who may enter the campus without authorization.

3. **Women Hostels and PG Accommodations** Women hostels and paying guest accommodations often require strong security systems to ensure the safety of residents. Traditional security systems rely mainly on guards or manual monitoring of CCTV cameras. By deploying the proposed IoT-based system, automatic face recognition can be used to identify authorized residents and detect unknown visitors. If an unfamiliar person enters the monitored area, the system captures their image and sends an alert. This helps hostel management maintain security and protect residents from potential threats.

4. Parking Areas and Basements, Parking lots and underground basements are often poorly monitored areas where security incidents may occur, especially at night. Women returning to their vehicles in such locations may face safety risks. Installing the proposed surveillance system in these areas allows continuous monitoring of vehicle parking zones. If an unfamiliar person loiters or enters the area, the system detects the face and sends an alert notification. This helps security personnel respond quickly and prevent suspicious activities.

5. Bus Stops and Public Transport Stations, Public transportation areas such as bus stops, railway stations, and metro entrances are locations where many people gather, especially during late hours. Women waiting alone in such areas may feel unsafe due to lack of monitoring. By installing this IoT-based monitoring system, authorities can automatically detect suspicious individuals and receive alerts. The captured image can help identify the person and assist in quick intervention if required.

6. Smart City Surveillance Systems, Many modern cities are implementing smart city technologies to improve public safety and infrastructure management. The proposed system can be integrated into smart city surveillance networks to provide intelligent monitoring capabilities. Using face recognition and automated alerts, city authorities can identify suspicious activities in real time. The system can also be connected to centralized control rooms where alerts from multiple locations can be monitored simultaneously.

CHAPTER-2

Literature Survey

2.1 Introduction

In our rapidly changing digital landscape, the definition of safety has shifted from physical locks and human guards to intelligent, data-driven systems. For a long time, surveillance was a "silent observer" cameras simply recorded events, and we only looked at the footage after something had already gone wrong. But for women navigating isolated streets or quiet campuses at night, a recording of an incident isn't enough; they need a system that can act before a situation escalates. This need for proactive safety is what drives the shift from passive recording to "Intelligent Perception."

The real challenge in modern security isn't just capturing video; it's overcoming "human latency." Human security guards, no matter how dedicated, face physical limits like exhaustion, boredom, and restricted visibility. Research now focuses on creating "Digital Watchdogs" that never sleep and never lose focus. By using accessible yet powerful technology like the Raspberry Pi, we can now place a high-level "brain" directly into a patrolling unit. This allows the system to monitor its surroundings with the same consistency every minute of the night.

Furthermore, the introduction of "Edge Intelligence" is a game-changer for privacy and speed. Instead of sending every face it sees to a distant cloud server, which takes time and risks data leaks, the "Ecliptic Watchdog" processes information locally, right where it happens. This means the system can recognize a familiar face or detect a stranger in milliseconds. When the system identifies a potential threat, it doesn't just wait; it uses the Internet of Things (IoT) to bridge the gap between the digital and physical worlds. By instantly sending a photo to a guardian's email, it transforms a piece of hardware into a reliable protector. This chapter explores the research and technical building blocks that allow us to turn this vision of an automated, intelligent guardian into a reality.

2.2 Evolution of Automated Surveillance Systems

The history of surveillance is a journey from simple human observation to complex machine intelligence. In its earliest forms, security was entirely dependent on the physical presence of a "watchman" or a patrolling officer. While this provided a human touch, it was inherently limited by the constraints of the human body: eyes can only see so far, and the mind can only stay alert for a few hours before fatigue sets in. The introduction of Closed-Circuit Television (CCTV) in the mid-20th century was the first major leap toward modernizing this process. It allowed a single person to monitor multiple locations from a central room. However, this still left the "Human Factor" as a weak link; studies have shown that after just twenty minutes of watching a monitor, a human operator's attention level drops significantly, often missing critical movements

The real transformation began with the digital revolution and the move toward **Active Surveillance**. Instead of being a passive tool that simply records video for later review, modern systems are being designed to "think" and "perceive." The transition from analog to digital IP cameras allowed for higher resolutions and, more importantly, the ability to integrate software directly into the video feed.

In recent years, the focus of research has shifted toward **Mobile Autonomous Surveillance**. Traditional cameras are fixed to walls or poles, creating "blind spots" that intruders can easily exploit. The evolution of robotics and embedded computing has now made it possible to create patrolling units that can navigate through environments. These mobile watchdogs act as a dynamic layer of security, constantly changing their vantage point and making it much harder for unauthorized individuals to avoid detection.

Today, we are in the era of **Intelligent Perception**, where AI and the Internet of Things (IoT) have merged. Surveillance is no longer just about "seeing"—it is about "understanding." Modern systems can now distinguish between a resident and a stranger, identify a cry for help, and instantly alert the right people with photographic evidence.

2.3 IoT-Based Smart Surveillance Systems

The Internet of Things (IoT) is an advanced technological paradigm that enables a wide range of physical devices to connect, communicate, and exchange data over the internet with minimal human intervention. These devices, embedded with sensors, software, and communication modules, can collect and share real-time information, making systems more intelligent and efficient. IoT has gained significant importance in various domains such as healthcare, agriculture, transportation, smart homes, and particularly in surveillance and security systems.

In IoT-based surveillance systems, multiple devices such as cameras, motion sensors, microcontrollers, and communication modules are interconnected through wired or wireless networks. These devices collaboratively monitor a specific environment, capture relevant data, and transmit it to a centralized system or cloud platform for processing and analysis. The integration of IoT with real-time monitoring capabilities allows continuous observation of an area without the need for constant human supervision, thereby improving efficiency and reliability.

In recent years, there has been a rapid increase in research and development of IoT-based safety and surveillance solutions. These systems are designed to detect unusual activities such as unauthorized access, motion in restricted areas, or suspicious behavior. Once such activities are identified, the system automatically generates alerts and sends notifications to users through various communication channels, including mobile applications, emails, SMS, or cloud-based dashboards. This real-time alert mechanism significantly reduces response time compared to conventional surveillance systems, where manual monitoring is required.

Furthermore, modern IoT surveillance systems often incorporate additional technologies such as computer vision, artificial intelligence, and machine learning to enhance their functionality. For instance, face recognition, object detection, and behavior analysis techniques enable the system to identify individuals and assess potential threats more accurately. This makes the surveillance system not only reactive but also predictive in nature.

2.4 Role of Raspberry Pi in Smart Systems

The Raspberry Pi is a small, affordable, and powerful computing device that has become very popular in the field of IoT and embedded systems. Despite its compact size, it is capable of performing many tasks similar to a regular computer. It supports various programming languages, especially Python, which makes it easy for students and developers to build and control different applications. One of the main advantages of Raspberry Pi is its ability to connect with a wide range of components such as cameras, sensors, motors, and other hardware devices, making it highly suitable for smart system development.

In many research projects and real-world applications, Raspberry Pi is commonly used for tasks like real-time image processing, data collection, and system control. For example, it can capture images or video through a camera module and process that data instantly to detect objects, motion, or other important activities. This makes it very useful in surveillance and security systems where quick decision-making is required.

Another important feature of Raspberry Pi is its support for edge computing. In this approach, data is processed locally on the device itself rather than being sent to cloud servers for processing. This reduces the delay (latency) in getting results and allows the system to respond much faster, which is especially important during emergency situations. It also reduces dependency on internet connectivity, making the system more reliable.

In the proposed system, Raspberry Pi plays a central role as the main processing unit. It is responsible for handling all major operations such as detecting activities, making decisions based on the data, and communicating alerts or information to the user. By combining its processing capability with connected sensors and cameras, the system becomes more efficient, intelligent, and capable of responding quickly to real-time events. This makes Raspberry Pi an ideal choice for building smart and effective surveillance systems.

2.5 Computer Vision and Face Recognition

Computer Vision is an important technology used in modern intelligent surveillance systems. It helps machines understand and analyze visual information such as images and videos, similar to how humans see and recognize things. By using computer vision, systems can identify objects, detect movements, and even recognize people, which makes surveillance more advanced and automated.

In many applications, computer vision is used to improve security by continuously monitoring an area and detecting any unusual activity. It reduces the need for constant human observation and allows the system to respond quickly when something suspicious is detected. This makes it a key component in building smart and efficient surveillance solutions.

One of the most commonly used tools for implementing computer vision is OpenCV. It is an open-source library that provides a wide range of functions for image processing and real-time vision applications. OpenCV is widely preferred because it is easy to use, flexible, and works well even on low-power devices like Raspberry Pi.

For face detection, techniques such as Haar Cascade classifiers are widely used. These methods are efficient and can quickly detect human faces in images or video streams. They are especially suitable for systems with limited processing power because they require less computational resources while still providing good accuracy.

For face recognition, algorithms like Local Binary Pattern Histogram (LBPH) are commonly used. This method works by analyzing the patterns in a person's face and comparing them with stored data. One of the main advantages of LBPH is that it performs well even under different lighting conditions, which is very important in real-world surveillance environments where lighting may not always be consistent.

2.6 Real-Time Alert Systems Using IoT

The Alert systems plays a very important role in modern safety and surveillance applications. The main purpose of these systems is to quickly inform users whenever any unusual or suspicious activity is detected. In earlier systems, alerts were limited and often required manual checking, but with the introduction of IoT, alert mechanisms have become faster, smarter, and more efficient.

Many existing surveillance solutions use different methods to send alerts, such as GSM modules, mobile applications, or cloud-based notification systems. These technologies allow users to receive messages or notifications on their smartphones or other devices, even if they are far away from the monitored location. This has greatly improved the ability to respond to emergencies in a timely manner.

One commonly used method for sending alerts is through SMTP-based email systems. These systems are popular because they are simple to implement and highly reliable. When an unusual event is detected, the system can automatically send an email to the user along with important details such as warning messages and captured images. This helps the user clearly understand the situation and take appropriate action quickly.

However, despite these advantages, many existing alert systems still have certain limitations. In some cases, alerts need to be triggered manually, which can delay the response time. Additionally, some systems do not have full automation and depend on user interaction to function properly. This reduces their effectiveness, especially in critical situations where immediate action is required.

To overcome these challenges, the proposed system introduces a more advanced and automated alert mechanism. It is designed to detect threats automatically without requiring any manual input. Once a suspicious activity is identified, the system instantly sends alerts along with captured images to the user. This ensures that the user is informed in real time and can respond quickly without any delay.

2.7 Hardware Interfacing and Mobile Surveillance

In traditional surveillance systems, cameras are usually fixed in one place, which limits their ability to cover larger areas. Because of this, there are always certain regions that remain unmonitored, reducing the overall effectiveness of the system. To overcome this limitation, recent research has focused on developing mobile surveillance systems, especially robotic platforms that can move around and monitor different locations dynamically.

Mobile surveillance robots are designed to patrol an area and continuously observe their surroundings. These robots use components like motor drivers to control their movement. One commonly used motor driver is the L293D, which allows the system to control DC motors and move the robot in different directions such as forward, backward, left, and right. This mobility helps in covering a wider area compared to fixed surveillance systems.

By integrating cameras and sensors with these mobile platforms, the system can capture real-time data while moving, making surveillance more flexible and efficient. These robots can be especially useful in large environments where installing multiple fixed cameras may not be practical.

However, many of the existing mobile surveillance systems still have certain drawbacks. Most of them focus only on movement and monitoring but lack intelligent decision-making capabilities. This means they may capture data but cannot effectively analyze it to detect threats or take immediate action.

To further enhance the effectiveness of mobile surveillance systems, the integration of intelligent technologies such as computer vision, artificial intelligence, and IoT connectivity is essential. By incorporating these advanced features, the system can not only monitor and move across different locations but also analyze the captured data in real time. For example, face recognition algorithms can be used to identify known and unknown individuals, while anomaly detection techniques can help in recognizing suspicious behavior.

2.8 Literature Summary and Gap Analysis

From the review of existing literature, it is clear that while many advancements have been made in surveillance and safety systems, several limitations still exist.

Firstly, many systems depend on manual activation, such as mobile safety applications where the user has to trigger an alert. This can be ineffective in emergency situations where the user may not be able to respond quickly.

Secondly, traditional CCTV systems are mostly fixed and require continuous human monitoring. This not only increases workload but also leads to possible errors due to lack of attention or delayed response.

In addition, some modern mobile surveillance systems have been developed to improve area coverage, but they often lack intelligent threat detection. They can move and capture data, but they are not capable of analysing the situation effectively. Another major limitation is that most existing systems do not provide a fully automated real-time response. They may detect an event, but the process of alerting or taking action is not completely automatic, which reduces their efficiency.

The proposed system, titled “Ecliptic Watchdogs,” is designed to overcome these challenges by introducing several important improvements. It provides fully automated surveillance, eliminating the need for user intervention. The system integrates computer vision techniques to enable intelligent threat detection, allowing it to identify suspicious activities accurately.

Another key advantage is the use of edge computing through Raspberry Pi, which allows data to be processed locally. This reduces delay and ensures a faster response, making the system more reliable in real-time situations.

Overall, the proposed system addresses the major gaps identified in existing research and provides a more advanced, efficient, and intelligent solution for modern surveillance needs.

CHAPTER-3

SYSTEM DESIGN AND COMPONENTS

3.1 Introduction

In recent years, surveillance systems have improved a lot, moving from simple monitoring systems to more intelligent and automated solutions. Earlier systems mainly depended on continuous human observation, but now technologies like IoT, Artificial Intelligence, and Computer Vision have made it possible to build smart systems that can detect and respond to threats on their own.

In this project, an intelligent surveillance system called **“Ecliptic Watchdogs”** is developed to provide real-time monitoring and protection, especially for women’s safety. The system is designed by combining different technologies such as a Raspberry Pi-based processing unit, a camera module, sensors, and a mobile robotic platform.

Unlike traditional CCTV systems that only record video, this system is capable of actively analysing the environment. It can detect the presence of a person, recognize faces, identify suspicious behaviour, and send alerts automatically without requiring human involvement. The system is also designed to be energy efficient and provide fast responses by using edge computing, where data is processed locally instead of relying on external servers

In addition to its core surveillance capabilities, the **“Ecliptic Watchdogs”** system is designed to ensure seamless communication and real-time alert generation. Whenever an unknown or suspicious individual is detected, the system immediately captures an image and sends notifications to predefined users through email or other IoT-based communication platforms. This instant alert mechanism helps in reducing response time and enables quick action to be taken by authorities or guardians. The system can also store captured data for future reference, which can be useful for investigation and analysis.

3.2 System Architecture

The overall system is divided into three main sections, each performing a specific function to ensure smooth operation.

The first part is the image acquisition unit, where a camera module is used to capture real-time video. The camera continuously monitors the surroundings and sends the captured frames to the processing unit for further analysis.

The second part is the processing unit, which is built using Raspberry Pi. This acts as the brain of the entire system. It processes the captured images using Python programming and OpenCV libraries. It performs important tasks such as detecting faces, identifying suspicious activities, and making decisions based on predefined conditions.

The third part is the alert and communication unit. This section is responsible for sending alerts to the user whenever a threat is detected. It uses IoT technology to send emails along with captured images as evidence. This ensures that the user receives immediate and clear information about the situation.

All these components work together in real time, which helps in reducing delay and improving the overall response efficiency of the system.

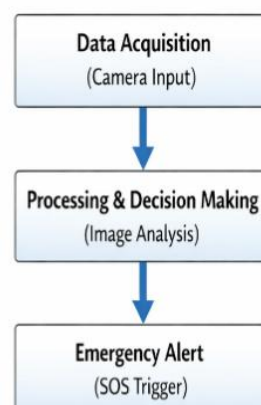


Fig 5: Flow Chart of System Architecture

3.3 Hardware Components

The proposed system makes use of several hardware components that work together to provide both intelligence and mobility. Each component plays an important role in ensuring that the system functions smoothly and performs surveillance tasks effectively.

The hardware setup is designed in such a way that all components are properly connected and communicate with each other without any delay. The Raspberry Pi acts as the central controller, while other components like the camera, motor driver, and motors support its operation. The camera is responsible for capturing real-time images and videos, which are then processed by the Raspberry Pi.

At the same time, the motor driver and DC motors help in providing movement to the system, allowing it to patrol different areas instead of staying in one fixed position. This combination of processing and mobility makes the system more flexible and efficient compared to traditional surveillance systems.

In addition, a proper power supply is provided to ensure continuous operation of all components. The integration of these hardware elements results in a reliable system capable of real-time monitoring, quick decision-making, and effective response.



(a) Raspberry Pi

Raspberry Pi

Raspberry Pi is a credit-card sized computer manufactured and designed in the United Kingdom by the Raspberry Pi foundation with the intention of teaching basic computer science to school students and every other person interested in computer hardware, programming and DIY-Do-it Yourself projects.

The Raspberry Pi is manufactured in three board configurations through licensed manufacturing deals with Newark element 14(Premier Farnell), RS Components and Egoman. These companies sell the Raspberry Pi online. Egoman produces a version for distribution solely in China and Taiwan, which can be distinguished from other Pi's by their red coloring and lack of FCC/CE marks. The hardware is the same across all manufacturers.

The Raspberry Pi has a Broadcom BCM2835 system on a chip (SoC), which includes an ARM1176JZF-S 700 MHz processor, Video Core IV GPU and was originally shipped with 256 megabytes of RAM, later upgraded (Model B & Model B+) to 512 MB. It does not include a built-in hard disk or solid-state drive, but it uses an SD card for booting and persistent storage, with the Model B+ using a MicroSD.

Hardware Layout

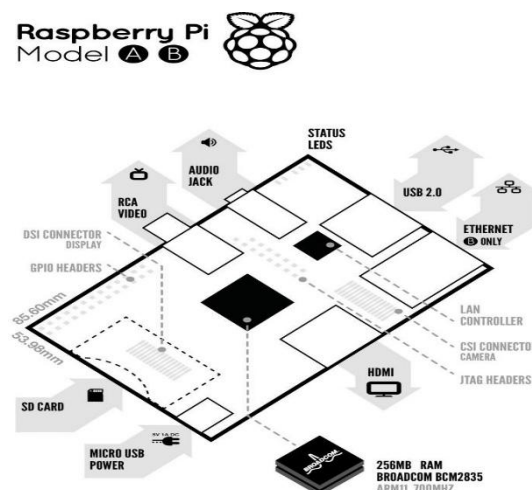


Fig 6: Hardware Layout of Raspberry Pi

DESCRIPTION OF THE COMPONENTS ON THE RASPBERRY PI:

1) Processor/SoC(System on Chip):

The Raspberry Pi has a Broadcom BCM2835 System on Chip module. It has a ARM1176JZF-S processor. The Broadcom SoC used in the Raspberry Pi is equivalent to a chip used in an old smartphone. While operating at 700 MHz by default, the Raspberry Pi provides a real world performance roughly equivalent to the 0.041GFLOPS. The Raspberry Pi chip operating at 700 MHz by default, will not become hot enough to need a heatsink or special cooling.

2) Power source:

The Pi is a device which consumes 700mA or 3W or power. It is powered by a Micro USB charger or the GPIO header. Any good smartphone charger will do the work of powering the Pi.

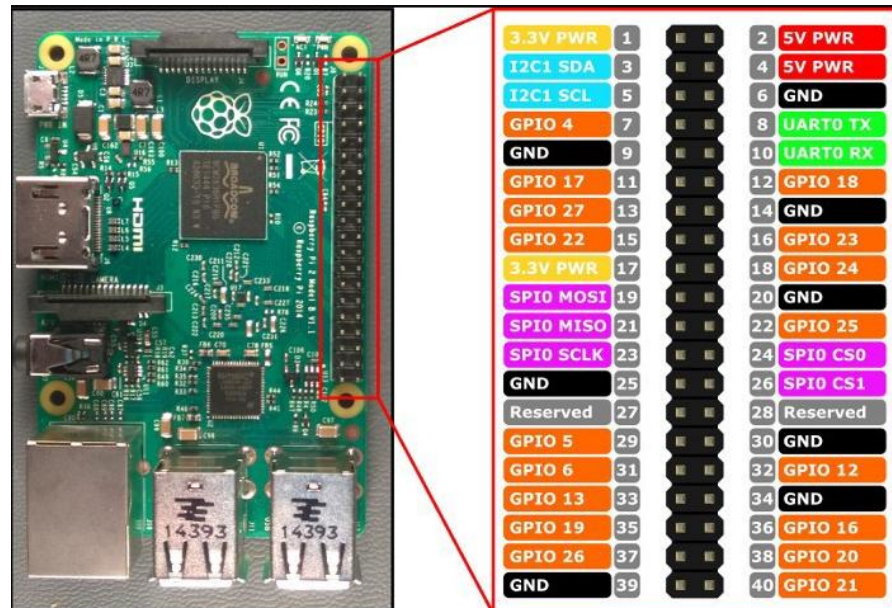
3) SD Card:

The Raspberry Pi does not have any onboard storage available. The operating system is loaded on a SD card which is inserted on the SD card slot on the Raspberry Pi. The operating system can be loaded on the card using a card reader on any computer.

4) GPIO:

General-purpose input/output (GPIO) is a generic pin on an integrated circuit whose behavior, including whether it is an input or output pin, can be controlled by the user at run time. GPIO pins have no special purpose defined, and go unused by default. The idea is that sometimes the system designer building a full system that uses the chip might find it useful to have a handful of additional digital control lines, and having these available from the chip can save the hassle of having to arrange additional circuitry.

The production Raspberry Pi board has a 26-pin 2.54 mm expansion header, marked as P1, arranged in a 2x13 strip. They provide 8 GPIO pins plus access to I²C, SPI, UART), as well as +3.3 V, +5 V and GND supply lines. Pin one is the pin in the first column and on the bottom row.



(b) Raspberry Pi Pin Configuration

5) DSI connector:

The Display Serial Interface (DSI) is a specification by the Mobile Industry Processor Interface (MIPI) Alliance aimed at reducing the cost of display controllers in a mobile device. It is commonly targeted at LCD and similar display technologies. It defines a serial bus and a communication protocol between the host and the device.

6) RCA Video:

RCA Video outputs (PAL and NTSC) are available on all models of Raspberry Pi. Any television or screen with a RCA jack can be connected with the RPi.



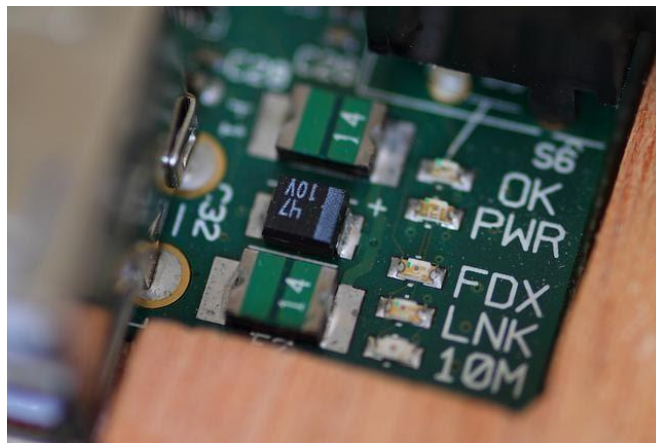
(c) Audio Jack

7) Audio Jack:

A standard 3.5 mm TRS connector is available on the RPi for stereo audio output. Any headphone or 3.5mm audio cable can be connected directly. Although this jack cannot be used for taking audio input, USB mics or USB sound cards can be used.

8) Status LEDs:

There are 5 status LEDs on the RPi that show the status of various activities. They are “OK”, “ACT”, “POWER” (PWR), Full Duplex (“FDX”), “LNK” (Link /Activity), “10M/100” which are shown in figure below.



(d) Status LEDs

9) USB 2.0 Port:

USB 2.0 ports are the means to connect accessories such as mouse or keyboard to the Raspberry Pi. There is 1 port on Model A, 2 on Model B and 4 on Model B+. The number of ports can be increased by using an external powered USB hub which is available as a standard Pi accessory.

10) Ethernet:

Ethernet port is available on Model B and B+. It can be connected to a network or internet using a standard LAN cable on the Ethernet port. The Ethernet ports are controlled by Microchip LAN9512 LAN controller chip.

11) CSI connector:

CSI – Camera Serial Interface is a serial interface designed by MIPI (Mobile Industry Processor Interface) alliance aimed at interfacing digital cameras with a mobile processor. The RPi foundation provides a camera specially made for the Pi which can be connected with the Pi using the CSI connector.

12) JTAG headers:

JTAG is an acronym for 'Joint Test Action Group', an organization that started back in the mid 1980's to address test point access issues on PCB with surface mount devices. The organization devised a method of access to device pins via a serial port that became known as the TAP (Test Access Port). In 1990 the method became a recognized international standard (IEEE Std 1149.1). Many thousands of devices now include this standardized port as a feature to allow test and design engineers to access pins.

13) HDMI:

HDMI –High-Definition Multimedia Interface

HDMI 1.3 a type A port is provided on the RPi to connect with HDMI screens.

SPECIFICATIONS:

	Model A	Model B	Model B+
Target price:	US\$25	US\$35	
SoC:	Broadcom BCM2835 (CPU, GPU, DSP, SDRAM, and single USB port)		
CPU:	700 MHz ARM1176JZF-S core (ARM11 family, ARMv6 instruction set)		
GPU:	Broadcom Video Core IV @ 250 MHz		
Memory (SDRAM):	256 MB GPU)	512 MB (shared with GPU)	

ECLIPTIC WATCHDOGS OF INTELLIGENT PERCEPTION, SMART IOT SURVEILLANCE FOR WOMEN PROTECTION

USB 2.0 ports:	1 (direct from BCM2835 chip) BCM2835 chip)	2 (via the on-board 3- port USB hub)	4 (via the on-board 5- port USB hub)
Video input:	15-pin MIPI camera interface (CSI) connector, used with the Raspberry Pi Camera Addon.		
Video outputs:	Composite RCA (PAL and NTSC) –in model B+ via 4-pole 3.5 mm jack, HDMI (rev 1.3 & 1.4), raw LCD Panels via DS		
Audio outputs:	3.5 mm jack, HDMI, and, as of revision 2 boards, I ² S audio (also potentially for audio input)		
Onboard storage:	SD / MMC / SDIO card slot (3.3 V card power support only) support only)	MicroSD	
Onboard network:	None	10/100 Mbit/s Ethernet (8P8C) USB adapter on the third/fifth port of the USB hub	
Low-level peripherals:	8× GPIO, UART, I ² C bus, SPI bus with two chip selects, I ² S audio +3.3 V, +5 V, ground		17× GPIO

Power ratings:	300 mA (1.5 W)	700 mA (3.5 W)	600 mA (3.0 W)
Power source:	5 V via Micro USB or GPIO header		
Size:	85.60 mm × 56 mm (3.370 in × 2.205 in) – not including protruding Connectors		
Weight:	45 g (1.6 oz)		

Table 1: Specifications

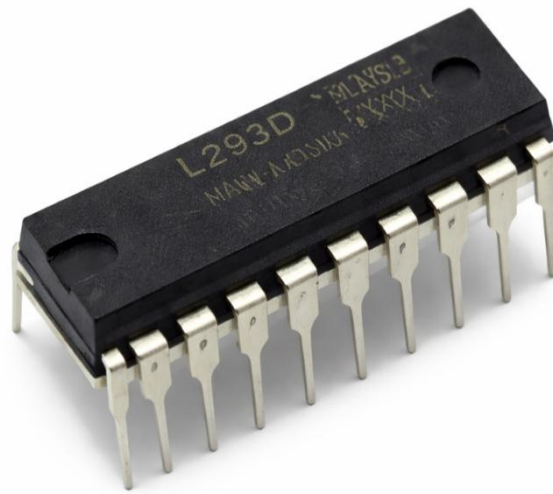


(e) Web Camera

Camera Module

The A **webcam** is a video camera that feeds or streams an image or video in real time to or through a computer network, such as the Internet. Webcams are typically small cameras that sit on a desk, attach to a user's monitor, or are built into the hardware. Webcams can be used during a video chat session involving two or more people, with conversations that include live audio and video.

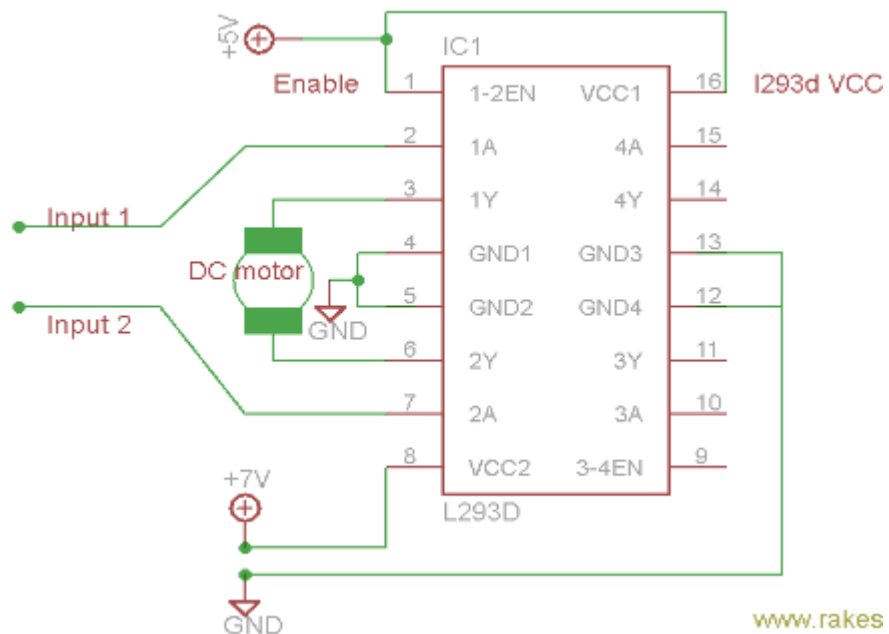
Webcam software enables users to record a video or stream the video on the Internet. As video streaming over the Internet requires much bandwidth, such streams usually use compressed formats. The maximum resolution of a webcam is also lower than most handheld video cameras, as higher resolutions would be reduced during transmission. The lower resolution enables webcams to be relatively inexpensive compared to most video cameras, but the effect is adequate for video chat sessions

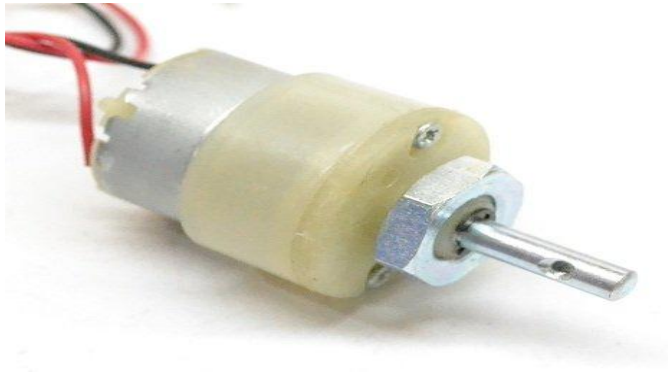


(f) L293D IC

Motor Driver (L293D)

The L293D motor driver is responsible for controlling the movement of the robot. It allows the system to move in multiple directions such as forward, backward, left, and right, making the surveillance system mobile and flexible.





(g) DC Motor

DC Motors

DC motors are used to provide movement to the robot. They convert electrical energy into mechanical motion, enabling the system to patrol different areas instead of remaining stationary.

The term ‘DC motor’ is used to refer to any rotary electrical machine that converts Direct current electrical energy into mechanical energy. DC motors can vary in size and Power from small motors in toys and appliances to large mechanisms that power vehicles, Pull elevators and hoists, and drive steel rolling mills.

When the coils are turned on and off in sequence, a rotating magnetic field is created that interacts with the differing fields of the stationary magnets in the stator to create torque, which causes it to rotate. These key operating principles of DC motors allow them to convert the electrical energy from direct current into mechanical energy through the rotating movement, which can then be used for the propulsion of objects.

IR sensor:

IR sensor is an electronic device, that emits the light in order to sense some object of the surroundings. An IR sensor can measure the heat of an object as well as detects the motion. Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation. These types of radiations are invisible to our eyes, but infrared sensor can detect these radiations.



(h) IR Sensor

The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode. Photodiode is sensitive to IR light of the same wavelength which is emitted by the IR LED. When IR light falls on the photodiode, the resistances and the output voltages will change in proportion to the magnitude of the IR light received.

There are five basic elements used in a typical infrared detection system: an infrared source, a transmission medium, optical component, infrared detectors or receivers and signal processing. Infrared lasers and Infrared LEDs of specific wavelength used as infrared sources.

The three main types of media used for infrared transmission are vacuum, atmosphere and optical fibers. Optical components are used to focus the infrared radiation or to limit the spectral response

Battery

A rechargeable battery is an energy storage device that can be charged again after being discharged by applying DC current to its terminals.

Rechargeable batteries allow for multiple usages from a cell, reducing waste and generally providing a better long-term investment in terms of dollars spent for usable device time.

A rechargeable battery is generally a more sensible and sustainable replacement to one-time use batteries, which generate current through a chemical reaction in which a reactive anode is consumed. The anode in a rechargeable battery gets consumed as well but at a slower rate, allowing for many charges and discharges.

Some types of batteries such as nickel cadmium and nickel-metal hydride can develop a battery memory effect when only partially discharged, reducing performance of subsequent charges and thus battery life in a given device.

Rechargeable batteries are used in many applications such as cars, all manner of consumer electronics and even off-grid and supplemental facility power storage.



(i) Battery

3.4 Software Requirements

PYTHON:

Python is a general purpose, dynamic, high level and interpreted programming language. It supports Object Oriented programming approach to develop applications. It is simple and easy to learn and provides lots of high-level data structures. It is easy to learn yet powerful and versatile scripting language which makes it attractive for Application Development. Its syntax and dynamic typing with its interpreted nature, makes it an ideal language for scripting and rapid application development. It supports multiple programming patterns, including object oriented, imperative and functional or procedural programming styles. It is not intended to work on special area such as web programming. That is why it is known as multipurpose because it can be used with web, enterprise, 3D CAD etc. We don't need to use data types to declare variable because it is dynamically typed so we can write `a=10` to assign an integer value in an integer variable. It makes the development and debugging fast because there is no compilation step included in python development and edit-test-debug cycle is very fast.

Advanced IP Scanner Desktop Application

The Advanced IP Scanner is a fast and user-friendly desktop application used for scanning and analyzing devices connected to a local network. It is widely used by network administrators, cybersecurity professionals, and IT support teams to monitor network activity, identify connected devices, and manage network resources efficiently.

This application works by scanning a specified range of IP addresses within a network and identifying all active devices. It uses network protocols such as ARP (Address Resolution Protocol), ICMP (Internet Control Message Protocol), and SNMP (Simple Network Management Protocol) to detect devices and gather relevant information such as IP address, MAC address, hostname, and device manufacturer.

One of the key features of Advanced IP Scanner is its high-speed scanning capability. The software uses multithreading techniques to scan large networks quickly and accurately, allowing users to discover hundreds or even thousands of devices

within seconds. This makes it highly efficient compared to traditional manual network monitoring methods.

The application also provides remote access and control functionalities. Users can remotely connect to devices using Remote Desktop Protocol (RDP) or integrated tools like Radmin.

Additionally, it allows users to perform operations such as remote shutdown and Wake-on-LAN, enabling efficient management of networked systems without physical access.

Another important feature is its ability to detect unauthorized or unknown devices connected to the network. This plays a crucial role in enhancing network security by helping administrators identify potential intrusions or suspicious devices in real time. The software also supports exporting scan results into formats such as CSV, which is useful for documentation and analysis purposes.

Advanced IP Scanner is designed with a simple and intuitive interface, making it easy to use even for beginners. It does not require complex configuration, and in many cases, it can be run as a portable application without installation. This portability makes it convenient for quick network diagnostics and troubleshooting.

However, despite its advantages, the tool has certain limitations. It lacks advanced features such as deep packet inspection, detailed port scanning, and automation capabilities, which are available in more sophisticated network analysis tools. Therefore, it is mainly suitable for basic network discovery and monitoring rather than advanced cybersecurity analysis.

In conclusion, Advanced IP Scanner is an efficient and lightweight desktop application for network scanning and device identification. Its speed, ease of use, and remote management features make it a valuable tool for maintaining network visibility and ensuring basic security, especially in small to medium-scale network environment.

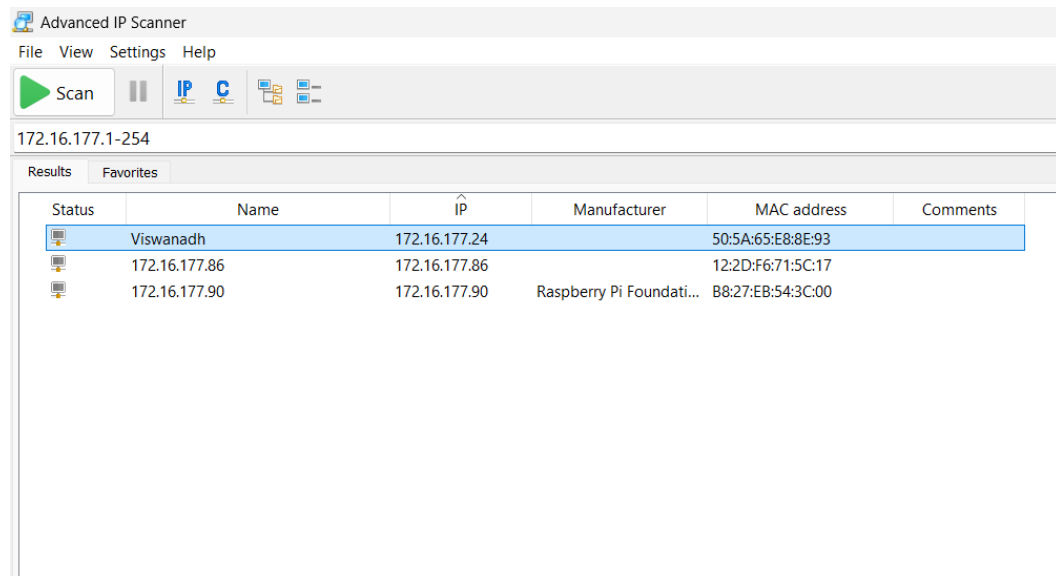


Fig 7: Advanced IP Scanner Interface

VNC Viewer (Virtual Network Computing Viewer)

VNC Viewer is a desktop application that allows users to remotely access and control another computer over a network. It is based on the Virtual Network Computing (VNC) technology, which uses the Remote Frame Buffer (RFB) protocol to transmit screen data from one system (server) to another (client).

VNC Viewer enables users to view the desktop of a remote computer and interact with it as if they were physically present. This includes controlling the mouse, typing using the keyboard, opening applications, and managing files. It is widely used in IT support, system administration, remote work, and educational environments.

The working of VNC Viewer involves two main components: the VNC Server and the VNC Viewer (client). The VNC Server is installed on the machine that needs to be accessed, while the VNC Viewer is installed on the device used to control it.

Once the connection is established using the IP address or hostname, the viewer displays the remote system's screen in real time. One of the key features of VNC Viewer is its cross-platform compatibility. It supports multiple operating systems such as Windows, Linux, macOS, Android, and iOS. This allows users to connect between

different types of devices seamlessly. Additionally, it supports secure connections using encryption and authentication mechanisms to protect data during transmission.

VNC Viewer also provides features such as file transfer, clipboard sharing, session recording, and multiple monitor support. These features enhance usability and make remote access more efficient and convenient. It is especially useful for troubleshooting systems, providing technical support, and managing servers remotely. Despite its advantages, VNC Viewer has some limitations. The performance may be affected by network speed, especially when accessing high-resolution desktops over slow connections. It also lacks some advanced features found in other remote desktop tools, such as optimized streaming or built-in collaboration tools in free versions.

In conclusion, VNC Viewer is a reliable and easy-to-use remote desktop application that enables efficient remote access and control of systems. Its simplicity, cross-platform support, and essential features make it a valuable tool for both personal and professional use.

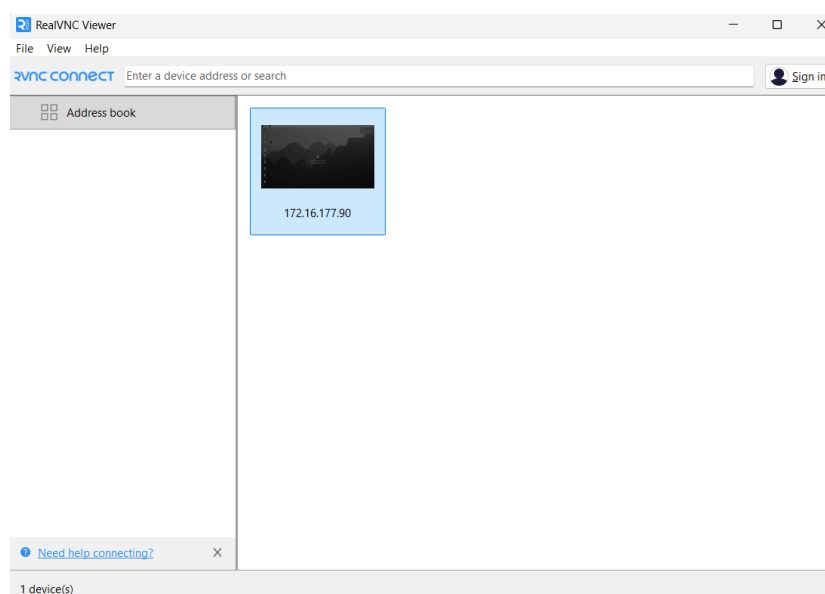


Fig 8 : VNC Viewer Interface

3.5 Software Implementation

The software implementation forms the core component of the proposed system, enabling intelligent monitoring, analysis, and automated decision-making. The system is developed using Python, a high-level programming language known for its simplicity, flexibility, and extensive library support. Python is widely used in computer vision and artificial intelligence applications due to its ease of integration with powerful libraries.

For real-time image processing and computer vision tasks, the system utilizes the OpenCV (Open Source Computer Vision Library). OpenCV provides a comprehensive set of tools and functions required for capturing video streams, processing images, detecting objects, and performing facial recognition. Its optimized performance and compatibility with embedded systems make it suitable for deployment on low-cost hardware platforms such as Raspberry Pi.

The software continuously captures live video input from the camera and processes each frame in real time. It performs multiple operations such as image conversion, face detection, recognition, and decision-making. The system is designed to operate autonomously, reducing the need for human intervention and ensuring continuous monitoring, especially during nighttime conditions.

Face Detection

Face detection is the first and most important step in the system, as it identifies the presence of human faces in the video stream. In this project, the Haar Cascade Classifier is used for detecting faces. This algorithm is based on machine learning and is trained using a large dataset of positive and negative images.

The Haar Cascade algorithm works by extracting Haar-like features from images and using a cascade of classifiers to detect objects efficiently. It scans the input image at multiple scales and identifies regions that resemble human facial features such as edges, lines, and textures. The use of an integral image allows the algorithm to compute these features quickly, making it highly efficient for real-time applications.

One of the major advantages of the Haar Cascade method is its speed and low computational requirement. This makes it ideal for systems running on limited hardware resources like Raspberry Pi. Despite being lightweight, it provides reasonably accurate results under normal lighting conditions and can detect multiple faces simultaneously.

In this system, once the face is detected, it is highlighted using a bounding box and passed to the next stage for recognition. The accuracy of face detection plays a critical role in the overall performance of the system, as incorrect detection can lead to false alerts or missed detections.

Face Recognition

After detecting the face, the system proceeds to identify the individual using a face recognition algorithm. In this project, the Local Binary Pattern Histogram (LBPH) algorithm is used for recognizing faces. LBPH is a widely used technique in real-time face recognition systems due to its simplicity and robustness.

The LBPH algorithm works by converting the facial image into a binary pattern based on pixel intensity comparisons. It divides the image into small regions and extracts local features by comparing each pixel with its neighboring pixels. These binary patterns are then converted into histograms, which represent the unique features of a face.

During the training phase, the system stores the facial data of known individuals along with their labels. When a new face is detected, its histogram is compared with the stored data to find the best match. If the match is within a predefined confidence level, the person is recognized as a known individual. Otherwise, the system classifies the person as unknown.

One of the key advantages of the LBPH algorithm is its ability to perform well under varying lighting conditions and facial expressions. Unlike some complex deep learning models, LBPH does not require high computational power, making it suitable

for embedded systems.

The recognition process ensures that only authorized or known individuals are identified correctly, while unknown persons are flagged for further action. This capability is essential for enhancing security and preventing unauthorized access.

Decision Making

The decision-making module is responsible for analyzing the results of face recognition and taking appropriate actions based on predefined conditions. This component adds intelligence to the system by enabling automated responses to potential threats.

When the system detects a face and recognizes it as a known individual, no action is taken, and the monitoring process continues normally. However, if the detected face is classified as unknown or suspicious, the system immediately initiates a series of actions.

Firstly, the system captures an image of the detected individual and stores it for future reference. This helps in maintaining a record of suspicious activities. Secondly, an alert is generated to notify the concerned authorities or users. The alert can be in the form of a notification, alarm, or message, depending on the system configuration.

The decision-making process is designed to operate in real time, ensuring that threats are identified and addressed immediately. This reduces the response time and increases the chances of preventing incidents before they occur.

Additionally, the system can be extended to include advanced features such as sending alerts via email or mobile applications, integrating with cloud storage, or triggering external devices like alarms and lights.

IoT-Based Alert System

The system uses IoT technology to enable real-time communication. SMTP protocol is used to send email alerts to the user whenever suspicious activity is detected. The captured images are attached as evidence, providing clear information about the situation.

This feature ensures a quick response during emergencies and allows users to monitor the system remotely from anywhere.

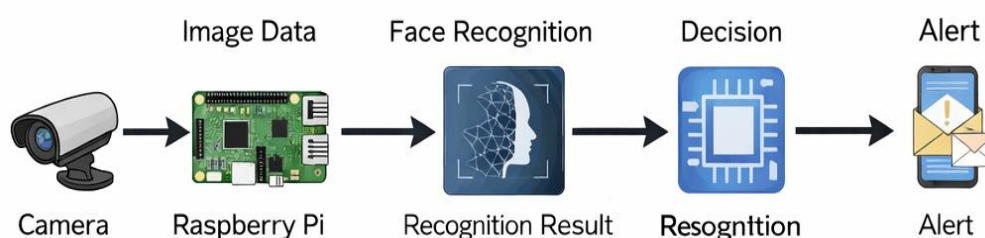


Fig 9 : Software Workflow

3.6 Mobile Surveillance Mechanism

Unlike traditional surveillance systems that remain fixed in one place, this project introduces a mobile surveillance robot. The robot is controlled using motor drivers and DC motors, allowing it to move and patrol different areas automatically.

This mobility helps the system cover a larger area and improves overall monitoring efficiency compared to static cameras.

3.7 System Workflow

The system operates in a continuous loop to ensure real-time monitoring. First, the camera captures live video, which is then processed by the Raspberry Pi. The system performs face detection and recognition on the captured frames.

If any suspicious activity or unknown person is detected, the system captures an image and sends an alert via email. Meanwhile, the robot continues its patrolling operation, ensuring uninterrupted surveillance.

3.8 Advantages of Proposed System

The proposed system offers several advantages:

1. It is fully automated and does not require manual activation
2. Provides real-time monitoring and instant alert system
3. Includes mobile surveillance capability
4. Cost-effective and energy efficient
5. Ensures fast response using edge computing

3.9 Conclusion

The Eclipse Watchdogs system presents an intelligent and effective solution for enhancing women's safety through the integration of IoT, computer vision, and robotic technologies. By overcoming the limitations of traditional surveillance systems, the proposed model introduces an automated and proactive approach to security monitoring.

The system ensures continuous real-time surveillance, enabling the detection of unauthorized individuals with high accuracy. Through the use of face recognition techniques, it can efficiently differentiate between known and unknown persons, thereby reducing false alarms and improving reliability. The automated alert mechanism further strengthens the system by providing immediate notifications during potential threats.

Despite its advantages, certain challenges such as performance in low-light conditions and dependency on dataset quality remain. These limitations can be addressed in future enhancements by incorporating advanced deep learning models and improved sensing technologies.

Overall, the proposed system offers a robust, reliable, and scalable solution for modern security requirements. It lays a strong foundation for the development of intelligent surveillance systems and contributes significantly toward building safer and smarter environments.

CHAPTER-4

Implementation

4.1 Database and Face Recognition Process

In this chapter, the database creation and face recognition workflow of the proposed system are explained in detail. The system relies on a structured dataset of facial images to accurately identify individuals and distinguish between known and unknown persons. The overall process involves dataset creation, input processing, face recognition, and output generation.

4.2 Dataset Creation

Dataset creation is a crucial step in building an effective face recognition system. It involves collecting multiple facial images of individuals who need to be recognized by the system. The dataset serves as the foundation for training the recognition model. In this system, facial images are captured using a camera module. For each person, multiple images are taken under different environmental conditions such as variations in lighting, background, facial expressions, and head orientations. This diversity helps the system learn a wide range of facial features, improving recognition accuracy in real-world scenarios.

Each individual is assigned a unique identifier (ID), which is used to label their images. The images are stored in a structured directory format, where each folder corresponds to a specific person. This organized storage makes it easier to retrieve and process the data during the training phase.

Before storing, the images undergo preprocessing steps such as:

- Conversion to grayscale (to reduce complexity)
- Resizing to a uniform dimension
- Noise reduction and normalization
- Face region extraction

These preprocessing techniques help in standardizing the dataset and improving the performance of the recognition algorithm.

A well-prepared dataset minimizes errors such as false positives and false negatives. Therefore, maintaining a balanced dataset is essential for achieving reliable results.

Name	Date modified	Type	Size
data	07-03-2026 14:23	File folder	
dataset	09-03-2026 08:20	File folder	
face_dataset	07-03-2026 14:31	Python File	2 KB
face_detect_VT	25-03-2026 16:19	Python File	5 KB
Intruder	25-03-2026 16:20	JPG File	67 KB
thief	13-07-2024 22:09	JPG File	81 KB
trainer.yml	10-04-2026 14:20	YML File	33,335 KB
training	06-12-2023 14:32	Python File	2 KB

Fig 10 : Data Set Created

4.3 Face Dataset Input and Training

After creating the dataset, the next step is to use it as input for training the face recognition model. During this phase, the system reads all stored images and processes them to extract meaningful features.

Each image is associated with its corresponding ID label. The system uses these labelled images to train the recognition algorithm, enabling it to differentiate between various individuals.

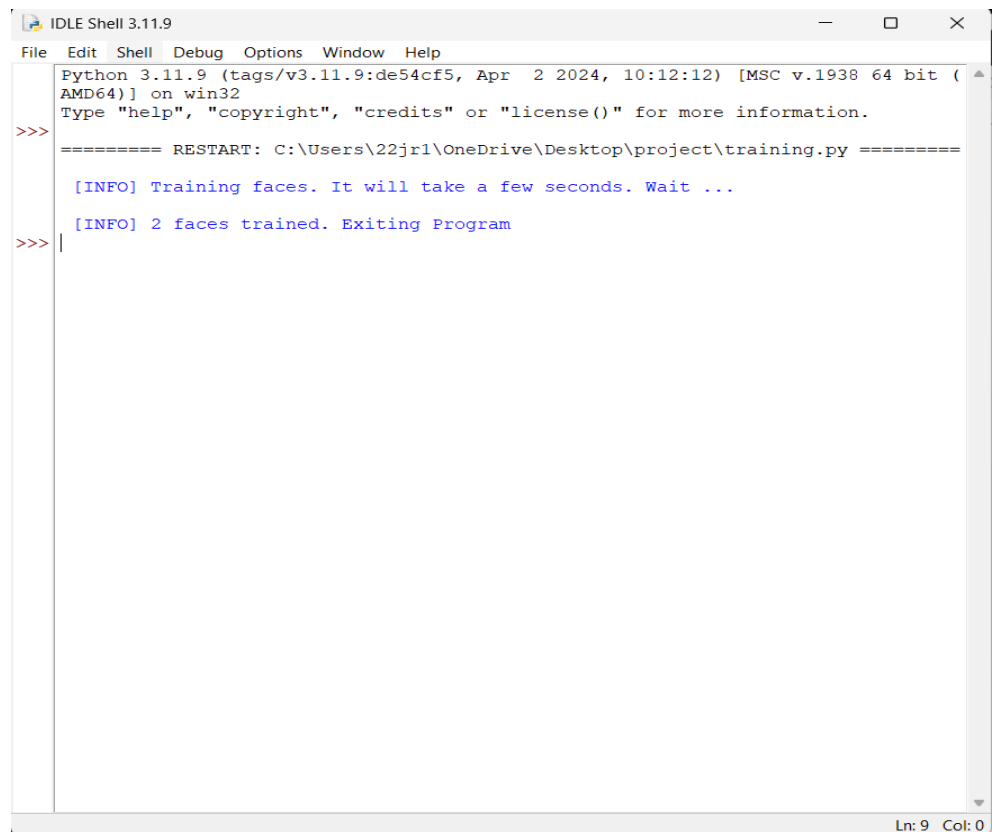
The training process involves:

1. Loading all images from the dataset
2. Detecting face regions within each image
3. Extracting features from the detected faces
4. Mapping extracted features to their respective IDs

The extracted features are then used to build a model that represents each individual's facial characteristics. This model is saved and later used for real-time recognition.

Training is a one-time or periodic process depending on whether new faces are added to the dataset. Whenever new individuals are introduced, the dataset is updated and the model is retrained to include the new data.

The efficiency of the training phase directly impacts the recognition accuracy. A well-trained model ensures faster and more precise identification during real-time operation.



```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\22jrl\OneDrive\Desktop\project\training.py =====
[INFO] Training faces. It will take a few seconds. Wait ...
[INFO] 2 faces trained. Exiting Program
>>> |
```

Fig 11: Data Set Training Completed

4.4 Face Recognition Process

The face recognition process is performed in real time using live video input from the camera. The system continuously captures frames and processes them to detect and recognize faces.

Initially, each frame is scanned to detect the presence of human faces using the face detection algorithm. Once a face is detected, the region of interest (ROI) is extracted and passed to the recognition module.

The recognition algorithm compares the detected face with the trained dataset by analysing its features. It calculates a confidence score that indicates how closely the detected face matches the stored images.

- If the confidence score is above a predefined threshold, the system identifies the person as a known individual.
- If the confidence score is below the threshold, the system labels the person as unknown.

This threshold-based classification helps in avoiding incorrect identification and ensures system reliability.

The recognition process is optimized for speed and efficiency, allowing the system to handle multiple faces simultaneously. This makes it suitable for real-time surveillance applications.

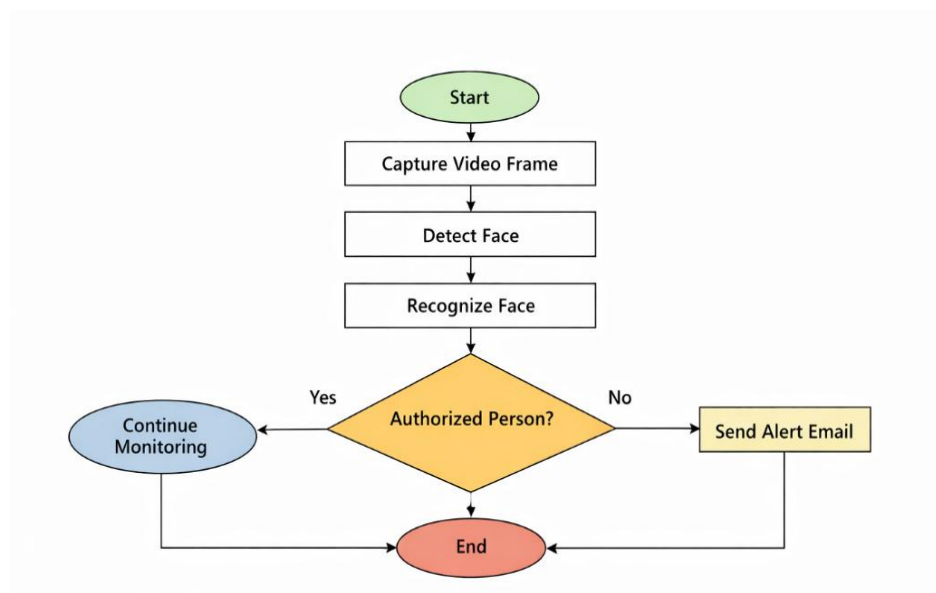


Fig 12 : Flow of data processing and decision-making in the system.

4.5 Output Generation

The output generation stage provides the final result of the recognition process. Based on

the classification of the detected face, the system generates appropriate outputs in real time.

For recognized individuals:

- The system displays the person's name or ID on the screen
- A bounding box is drawn around the detected face
- The identification result is updated continuously

For unknown individuals:

- The system labels the person as "Unknown"
- An alert may be triggered (such as a notification or alarm)
- The image of the unknown person may be stored for future analysis

Additionally, the system can maintain logs of recognized and unrecognized individuals.

These logs can be useful for security audits and system improvements.

The output is designed to be clear and immediate, enabling quick decision-making and response. This real-time feedback significantly enhances the effectiveness of the surveillance system.

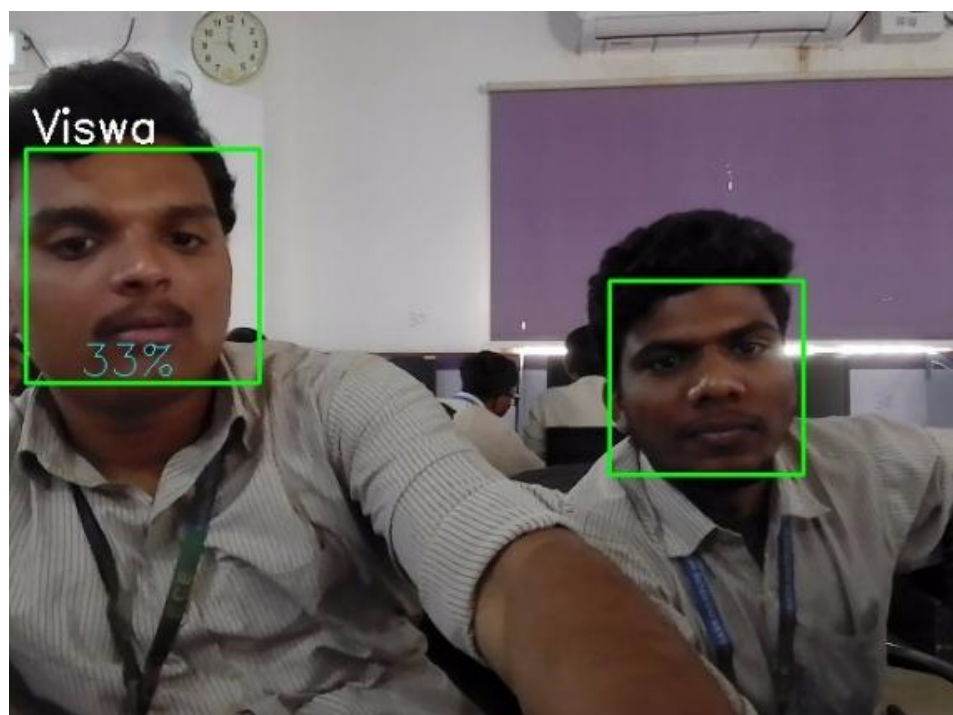


Fig 13 : Face detection

4.6 Hardware Integration and Prototype Design

This section explains the hardware connections and the final prototype design of the proposed surveillance system. The integration of hardware components with the software modules plays a crucial role in achieving real-time monitoring and face recognition.

The system is built using a camera module connected to a processing unit such as a Raspberry Pi or a computer.

In addition to the camera and processing unit, the system may include output devices such as a display screen and alert mechanisms. The display is used to show real-time results, including detected faces, recognized identities, and system status. Alert systems such as buzzers or notification modules can be integrated to provide immediate warnings when an unknown person is detected. The hardware components are connected in a structured manner to ensure smooth data flow.

Overall, the hardware integration ensures that the system operates effectively by combining physical components with intelligent software. The successful development of the prototype proves that the system can be used in real-world surveillance applications to enhance safety and security.



Fig 14 : Hardware Kit of The Proposed Project

CHAPTER-5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the implementation of the proposed surveillance system based on computer vision and face recognition techniques. The main objective of this chapter is to evaluate the performance of the system under different conditions and analyse its effectiveness in real-time monitoring applications.

The system was tested using live video input, where facial images were captured, processed, and analysed continuously. The performance of each module, including dataset utilization, face detection, face recognition, and decision-making, is examined in detail. The results demonstrate how efficiently the system can identify individuals and classify them as known or unknown.

In addition to presenting the results, this chapter also provides a detailed discussion of system behaviour under various scenarios such as different lighting conditions, multiple face detection, and variations in dataset quality. The strengths and limitations of the system are analysed to understand its reliability and practical applicability.

Furthermore, performance metrics such as accuracy, speed, and response time are evaluated to determine the overall efficiency of the system. The outputs generated by the system, including recognized faces, unknown face detection, and alert mechanisms, are also discussed.

This chapter plays a crucial role in validating the proposed system by demonstrating its real-time capabilities and effectiveness in enhancing surveillance and security. The findings presented here provide a clear understanding of how the system performs in real-world conditions and highlight areas for future improvement.

5.2 System Implementation Results

The proposed surveillance system was successfully implemented using Python programming language along with the OpenCV library for real-time image processing and analysis. The system integrates multiple modules such as video capture, face detection, face recognition, and decision-making into a single framework. Each module works in coordination to provide continuous monitoring and intelligent response.

During implementation, a camera module was used to capture live video streams. The system processes each frame in real time, ensuring that there is no significant delay between input and output. The integration of Haar Cascade for face detection and LBPH algorithm for face recognition proved to be efficient and suitable for real-time applications.

The results show that the system is capable of detecting faces accurately and recognizing individuals based on the trained dataset. Known individuals are identified and labelled correctly, while unknown individuals are marked as “Unknown.” The system also successfully triggers alerts when unfamiliar faces are detected.

Overall, the implementation results confirm that the system meets the objectives of real-time surveillance and automated face recognition with good performance and reliability.

5.3 Dataset Evaluation Results

The dataset plays a crucial role in determining the accuracy and performance of the face recognition system. In this project, the dataset was created by capturing multiple facial images of individuals under different environmental conditions. The dataset was evaluated to analyse its impact on the system's overall performance.

During dataset preparation, images were collected with variations in lighting, facial expressions, angles, and background conditions. These variations are important because they help the system learn diverse facial features, making it more robust during real-time recognition. Each image was labelled with a unique ID corresponding to a specific individual, and the dataset was organized in a structured format.

The evaluation results indicate that the size and quality of the dataset significantly affect recognition accuracy. When the dataset contained a higher number of images per individual, the system showed improved performance. This is because the recognition model was able to learn more variations of the same face, resulting in better matching during real-time processing.

It was also observed that preprocessing steps such as grayscale conversion, resizing, and normalization helped in improving the consistency of the dataset

These steps reduced noise and unnecessary variations, allowing the recognition algorithm to focus on important facial features. However, certain limitations were identified during dataset evaluation. Images with poor lighting, blur, or improper face alignment negatively affected the system's performance. In cases where the dataset contained very few images of an individual, the system struggled to recognize them accurately. This highlights the importance of maintaining a well-balanced and high-quality dataset.

Additionally, the dataset was tested for its ability to handle unknown individuals. The system was able to correctly classify faces not present in the dataset as "unknown," which is an important feature for security applications.

Overall, the dataset evaluation results confirm that a well-structured and diverse dataset is essential for achieving high accuracy in face recognition systems. Proper dataset preparation

and maintenance play a key role in ensuring reliable system performance.

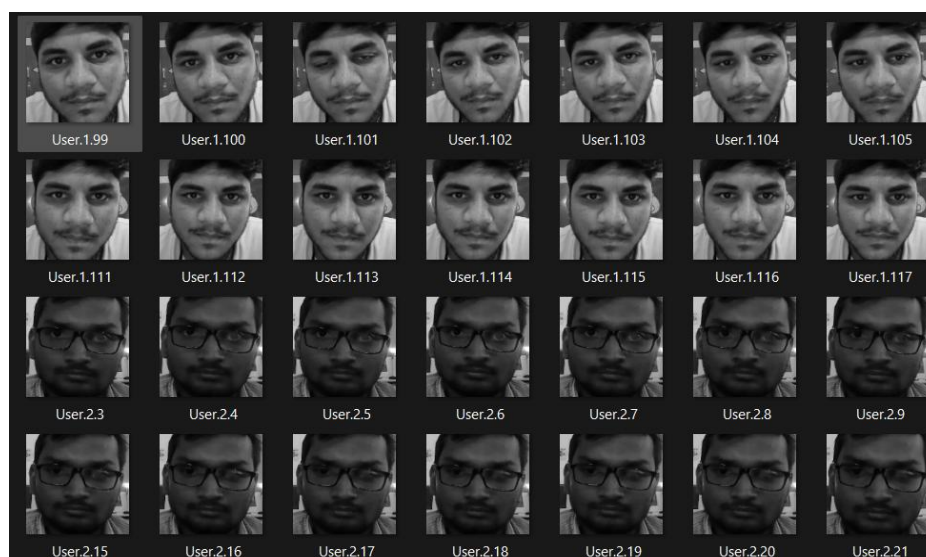


Fig 15: Data Set Created for Users

5.4 Face Detection Results

The face detection module is a fundamental component of the proposed surveillance system, as it is responsible for identifying the presence and location of human faces in the video stream. In this project, the Haar Cascade classifier was used for face detection due to its efficiency, speed, and suitability for real-time applications.

The performance of the face detection system was evaluated under different environmental and operational conditions. These included variations in lighting, the presence of multiple individuals, different distances from the camera, and changes in facial orientation. The system was tested using live video input, where each frame was processed continuously to detect faces.

The results showed that the system was able to detect faces accurately in normal lighting conditions. The Haar Cascade algorithm successfully identified facial regions and drew bounding boxes around detected faces in real time. The detection speed was high, allowing the system to process multiple frames per second without noticeable delay. This confirms that the algorithm is well-suited for real-time surveillance applications.

In scenarios involving multiple individuals, the system demonstrated the ability to detect more than one face simultaneously. Each detected face was marked separately, indicating that the system can handle real-world situations such as crowded environments. This capability is essential for surveillance systems where multiple people are at the same time.

However, certain limitations were observed during testing. In low-light conditions, the detection accuracy decreased slightly, and some faces were not detected properly. Similarly, when faces were partially covered or when individuals were positioned at extreme angles, the system occasionally failed to detect them. These limitations are common in traditional face detection techniques and highlight the need for improved algorithms in challenging conditions.

Despite these challenges, the overall performance of the face detection module was satisfactory. Its lightweight nature ensures low computational requirements, making it suitable for deployment on devices like Raspberry Pi. The balance between speed and accuracy makes Haar Cascade an effective choice for this system.

In conclusion, the face detection module successfully achieved its objective of identifying human faces in real time. It provides a reliable foundation for the face recognition module, ensuring that only relevant facial regions are processed for identification.

5.5 System Performance Analysis

The performance of the proposed surveillance system was analyzed based on several key parameters, including accuracy, speed, reliability, and scalability. This analysis helps in evaluating how effectively the system performs under real-time conditions and different environmental factors.

Accuracy

Accuracy is one of the most important parameters in a face recognition system. It refers to the system's ability to correctly identify known individuals and classify unknown persons. The system achieved high accuracy when tested with a well-trained dataset containing multiple images per individual. The use of preprocessing techniques and the LBPH algorithm contributed to improved recognition performance.

Speed

The system demonstrated real-time processing capability by analyzing video frames continuously without noticeable delay. The use of lightweight algorithms such as Haar Cascade for detection and LBPH for recognition ensured fast execution.

The system was able to process multiple frames per second, making it suitable for live surveillance applications. Fast processing speed is essential for timely detection and response to potential threats.

Reliability

Reliability refers to the system's ability to perform consistently over time. The system showed stable performance under normal conditions and was able to continuously detect and recognize faces without interruptions.

However, reliability decreased slightly in extreme conditions such as very low light or when faces were partially covered. These factors may lead to occasional detection or recognition errors.

Scalability

The system is scalable and can be extended by adding more individuals to the dataset. The model can be retrained with new data, allowing the system to adapt to changing requirements.

This flexibility makes the system suitable for deployment in different environments such as homes, offices, and public areas.

Parameter	Observation	Result
Detection Accuracy	High in normal lighting	Good
Recognition Accuracy	Depends on dataset quality	Satisfactory
Processing Speed	Real-time performance	Fast
Reliability	Stable under normal use	Moderate
Scalability	Easily extendable	High

Table 2: Performance of the System

5.6 Output Results

The output results of the proposed system demonstrate its ability to perform real-time face detection and recognition effectively. The system provides visual and informational outputs that allow users to monitor and analyze activities continuously.

During operation, the system captures live video input and processes each frame to detect faces. Once a face is detected, a bounding box is drawn around the face to highlight the detected region. This helps in clearly identifying the presence of individuals in the monitored area.

When the detected face matches with the trained dataset, the system recognizes the individual and displays their corresponding name or ID on the screen. This information is shown along with the bounding box, providing clear identification of authorized or known persons.

The recognition results are updated continuously in real time, ensuring accurate monitoring.

In cases where the detected face does not match any stored data, the system labels the individual as “Unknown.” This is a critical feature for security applications, as it helps in identifying unauthorized or suspicious persons. Upon detecting an unknown individual, the system can automatically capture the image and store it for future reference.

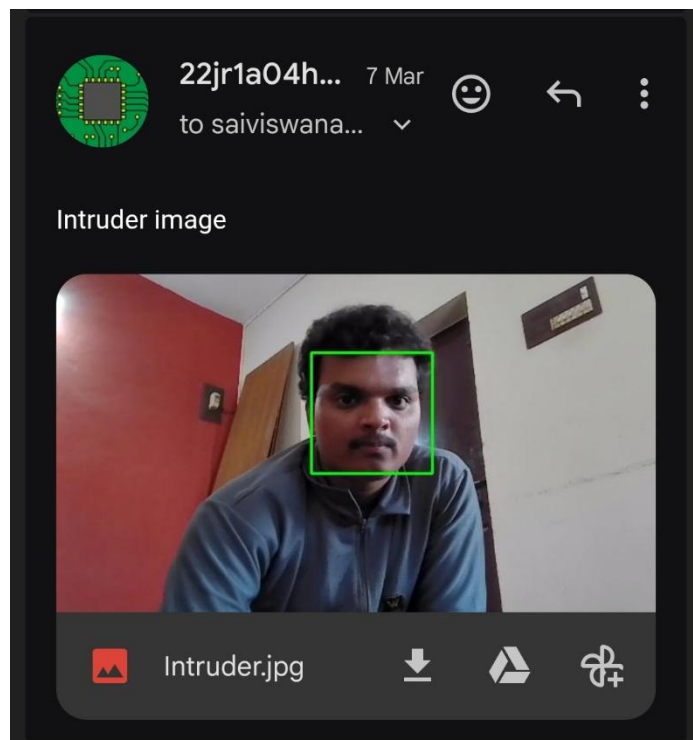


Fig 16: Unknown Person Recognized as Intruder

Additionally, the system is capable of generating alerts when unknown faces are detected. These alerts may include visual notifications, warning messages, or triggering external devices such as buzzers, depending on the system configuration. This ensures a quick response to potential threats and enhances the overall effectiveness of the surveillance system.

The output results are displayed in real time with minimal delay, making the system highly responsive. The clarity of the output, including bounding boxes and labels, ensures ease of understanding for the user. The system interface is simple and user-friendly, allowing efficient monitoring without requiring technical expertise. To validate the performance of the

system, different test cases were considered. These include recognizing known individuals, detecting unknown persons, and handling multiple faces simultaneously. The system successfully handled these scenarios and produced accurate outputs in most cases.

Including output screenshots in the report further strengthens the validation of the system. Typical outputs include:

- Face detection with bounding box
- Recognized face with name/ID
- Unknown face labeled clearly
- Alert notification display
-

In conclusion, the output results confirm that the system performs its intended functions effectively. It provides real-time monitoring, accurate face recognition, and immediate alert generation, making it a reliable solution for surveillance and security applications.

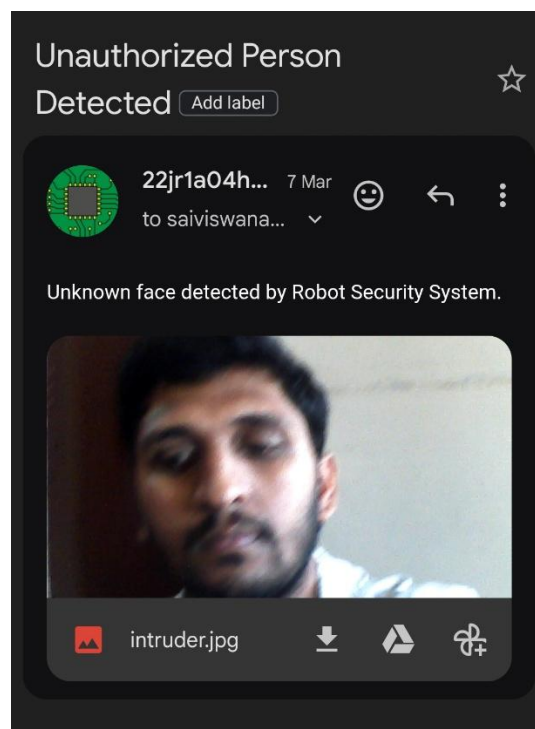


Fig 17: Email Alert Sent After Detecting Unknown Person Detected

CHAPTER-6

Conclusion and Future Scope

Conclusion And Future Scope

In this proposed Ecliptic Watchdogs system presents an efficient and intelligent solution for enhancing safety through real-time surveillance. By integrating IoT, computer vision, and automated alert mechanisms, the system overcomes the limitations of traditional monitoring methods. It ensures continuous observation, accurate detection of unauthorized individuals, and immediate notification, thereby improving response time and overall security.

The system demonstrates reliable performance under normal conditions, with high detection and recognition accuracy. The use of Raspberry Pi and OpenCV enables a cost-effective and scalable implementation suitable for real-world applications such as campuses, residential areas, and public spaces. The modular design allows easy integration of additional features, making the system adaptable to different requirements.

However, certain limitations such as reduced performance in low-light environments and dependency on dataset quality were observed. These challenges highlight the need for further improvements to enhance system robustness and accuracy.

Future work can focus on integrating advanced deep learning techniques to improve recognition accuracy under varying conditions. The system can also be enhanced by incorporating mobile application support for real-time notifications, cloud storage for data management, and additional sensors for improved monitoring. Furthermore, expanding the system into a fully autonomous security network can significantly contribute to smart city development.

In conclusion, the proposed system provides a reliable, scalable, and cost-effective approach for modern surveillance and safety applications, laying the foundation for future intelligent security solutions.

CHAPTER-7

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Appendix-1: Face Data-Set Code

```
import cv2
import sys
import os
import time

cam = cv2.VideoCapture(0)
cam.set(3, 640) # set video width
cam.set(4, 480) # set video height

face_detector =
cv2.CascadeClassifier('C:/Users/22jr1/OneDrive/Desktop/project/data/haarcascade_fr
ontalface_default.xml')

# For each person, enter one numeric face id
face_id = input("\n enter user id end press <return> ==> ")

print("\n [INFO] Initializing face capture. Look the camera and wait ...")

# Initialize individual sampling face count
count = 0

while(True):
    ret, img = cam.read()
    #img = cv2.flip(img, -1) # flip video image vertically
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    faces = face_detector.detectMultiScale(gray, 1.3, 5)

    for (x,y,w,h) in faces:
```

```

cv2.rectangle(img, (x,y), (x+w,y+h), (255,0,0), 2)
count += 1

# Save the captured image into the datasets folder
cv2.imwrite("dataset/User." + str(face_id) + '.' + str(count) + ".jpg",
gray[y:y+h,x:x+w])

cv2.imshow('image', img)

# Press 'ESC' for exiting video
if cv2.waitKey(100) & 0xff == ord('q'):
    break
elif count >= 120: # Take 80 face sample and stop video
    break

print("\n [INFO] Exiting Program and cleanup stuff")
cam.release()
cv2.destroyAllWindows()

```

Appendix-2: Data-Set Training Code

```

import cv2
import numpy as np
from PIL import Image
import os

# Path for face image database
path = 'dataset'

recognizer = cv2.face.LBPHFaceRecognizer_create()
detector = cv2.CascadeClassifier("data/haarcascade_frontalface_default.xml");

```

```

# function to get the images and label data
def getImagesAndLabels(path):
    imagePath = [os.path.join(path,f) for f in os.listdir(path)]
    faceSamples=[]
    ids = []
    for imagePath in imagePath:
        PIL_img = Image.open(imagePath).convert('L') # convert it to grayscale
        img_numpy = np.array(PIL_img,'uint8')
        id = int(os.path.split(imagePath)[-1].split(".")[1])
        faces = detector.detectMultiScale(img_numpy)
        for (x,y,w,h) in faces:
            faceSamples.append(img_numpy[y:y+h,x:x+w])
            ids.append(id)
    return faceSamples,ids

print ("\n [INFO] Training faces. It will take a few seconds. Wait ...")
faces,ids = getImagesAndLabels(path)
recognizer.train(faces, np.array(ids))

# Save the model into trainer/trainer.yml
recognizer.save('trainer.yml') # recognizer.save() worked on Mac, but not on Pi

# Print the numer of faces trained and end program
print("\n [INFO] {0} faces trained. Exiting Program".format(len(np.unique(ids))))

```

Appendix-3: Face Detection Code

```

import smtplib
from email.mime.text import MIMEText#email.mime.text.MIMEText(_text[,
_subtype[, _charset]])
from email.mime.multipart import MIMEMultipart
from email.mime.base import
MIMEBase#email.mime.base.MIMEBase(_maintype(e.g. text or image),
_subtype(e.g. plain or gif), **_params(e.g.key/value dictionary))
from email import encoders
import os.path
import cv2
import numpy as np
import time
##import os
from time import sleep
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.read('trainer.yml')
cascadePath = "data/haarcascade_frontalface_default.xml"
faceCascade = cv2.CascadeClassifier(cascadePath);
font = cv2.FONT_HERSHEY_SIMPLEX
#iniciate id counter
id = 0
# names related to ids: example ==> Marcelo: id=1, etc
names = ['None', 'user1', 'user2', 'user3', 'user4', 'user5']
# Initialize and start realtime video capture
cam = cv2.VideoCapture(0)
##cam.set(3, 640) # set video width
##cam.set(4, 480) # set video height
# Define min window size to be recognized as a face
i=0

```

count=1 [Ecliptic Watchdogs of Intelligent Perception - Paper Publication.docx](#)

```
def sendmail():
    email = '22jr1a04h2@gmail.com'
    password = 'ikkfvhddfdmkrdvc'
    send_to_email = 'saiviswanadh7@gmail.com'
    subject = 'Final year project'
    message = 'Intruder image'
    file_location = 'C:/Users/22jr1/OneDrive/Desktop/project/Intruder.jpg'
    #/home/pi/Downloads/takeoff.png
    msg = MIMEMultipart()#Create the container (outer) email message.
    msg['From'] = email
    msg['To'] = send_to_email
    msg['Subject'] = subject
    """as.string()
    |
    +-----MIMEMultipart
            |                               |---content-type
            |                               +---header---+---content disposition
            +----.attach()-----+----MIMEBase----|
                                |       +---payload (to be encoded in Base64)
                                +----MIMEText"""
    msg.attach(MIMEText(message, 'plain'))#attach new message by using the
    Message.attach
    filename = os.path.basename(file_location)#function returns the tail of the path
    attachment = open(file_location, "rb") #(read binary)
    part = MIMEBase('application', 'octet-stream')#Content-Type: application/octet-
    stream , image/png, application/pdf
    part.set_payload((attachment).read())
    encoders.encode_base64(part)
    part.add_header('Content-Disposition', "attachment; filename= %s" %
    filename)#Content-Disposition: attachment; filename="takeoff.png"
    msg.attach(part)
```

```

server = smtplib.SMTP('smtp.gmail.com', 587)# Send the message via local SMTP
server.

server.starttls()# sendmail function takes 3 arguments: sender's address, recipient's
address and message to send

server.login(email, password)
text = msg.as_string()
server.sendmail(email, send_to_email, text)
server.quit()
print("Mail sent");

while True:
    ret, image =cam.read()
    #img = cv2.flip(img, -1) # Flip vertically
    gray = cv2.cvtColor(image,cv2.COLOR_BGR2GRAY)
    faces = faceCascade.detectMultiScale(
        gray,
        scaleFactor = 1.2,
        minNeighbors = 5
    )
    for(x,y,w,h) in faces:
        cv2.rectangle(image, (x,y), (x+w,y+h), (0,255,0), 2)
        id, confidence = recognizer.predict(gray[y:y+h,x:x+w])
        # Check if confidence is less them 100 ==> "0" is perfect match
        if (confidence < 90):
            id = names[id]
            confidence = " {0}%".format(round(100 - confidence))
            if(count == 1):
                print('face id:',str(id),' detected')
                count =2
                i=0
            else:
                id = "unknown"

```

```

print('no face detected')
confidence = " {0}%".format(round(100 - confidence))
time.sleep(0.1)
cv2.imwrite(filename='Intruder.jpg', img=image)
print("Image captured")
sendmail()
time.sleep(1)
cv2.putText(image, str(id), (x+5,y-5), font, 1, (255,255,255), 2)
cv2.putText(image, str(confidence), (x+5,y+h-5), font, 1, (255,255,0), 1)
cv2.imshow('camera',image)
i= i+1
#print('i: ',int(i))
if(i == 100):
    #print('i: ',int(i))
    count=1
    i=0
k = cv2.waitKey(10) & 0xff # Press 'ESC' for exiting video
if k == 27:
    break
# Do a bit of cleanup
print("\n [INFO] Exiting Program and cleanup stuff")
cam.release()
cv2.destroyAllWindows()

```

Ecliptic Watchdogs of Intelligent Perception: Smart IoT Surveillance for Women Protection

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Abstract: The Ecliptic Watchdog is a smart IoT-based surveillance system designed to enhance women's safety. Unlike traditional systems that only record, it actively detects threats using real-time sensors and intelligent algorithms. It triggers immediate SOS alerts in case of potential danger. This project focuses on creating a miniaturized, efficient node that communicates reliably in various environments, making it a potent tool for protection. The system shifts from passive recording to active threat detection, with real-time sensors playing a key role. Intelligent perception algorithms enable smart decision-making, ensuring immediate alerts for help. Overall, the Ecliptic Watchdog aims to provide enhanced safety and security for women, making it an efficient and reliable solution. The system's design allows it to work effectively in diverse settings. Its advanced features make it a valuable tool for women's protection. The Ecliptic Watchdog represents a significant step forward in safety tech.

Keywords: Smart IoT Surveillance, Women's Safety, Real-Time Threat Detection, Intelligent Perception Algorithms, SOS Alert System, Raspberry Pi, Python Programming, Motor Drivers, DC Motors, Camera Module, Embedded Systems, Wireless Communication.

I. Introduction

The rapid growth of smart technologies, Internet of Things (IoT), and real-time monitoring systems has significantly increased the demand for intelligent security solutions. In recent years, women's safety has become a major social concern, requiring proactive and technology-driven protection systems. Traditional surveillance systems mainly focus on passive video recording, where incidents are reviewed only after they occur. Such systems suffer from delayed response and lack automated threat detection capabilities. These limitations have encouraged the development of smart IoT-based surveillance systems capable of real-time monitoring and immediate emergency response.

IoT-based security systems provide significant advantages, including remote monitoring, real-time data processing, automated alerts, and improved reliability. By integrating sensors, embedded processors, and wireless communication technologies, modern surveillance systems can actively detect suspicious activities instead of merely recording them. This shift from passive observation to intelligent perception plays a crucial role in enhancing personal safety and emergency response efficiency.

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The Ecliptic Watchdog is an advanced smart surveillance system designed specifically to enhance women's safety. The system is developed using a Raspberry Pi as the core processing unit and programmed using Python for real-time data analysis and system control. It integrates a camera module for continuous monitoring, along with motor drivers and DC motors to enable controlled movement and wider environmental coverage. This hardware integration ensures efficient surveillance across diverse surroundings.

Unlike conventional monitoring systems, the proposed system performs active threat detection using intelligent perception algorithms. Real-time image processing and sensor data analysis allow the system to identify unusual or potentially dangerous situations. When a threat is detected, the system immediately triggers an SOS alert to predefined contacts, ensuring quick assistance and improved emergency response.

The Ecliptic Watchdog represents a significant advancement in smart safety technology by combining IoT, embedded systems, real-time monitoring, and intelligent decision-making. Its ability to function effectively in diverse environments makes it a practical, scalable, and reliable solution for women's protection.

IoT-based intelligent surveillance systems have become widely used due to their ability to provide real-time monitoring, motion detection, and automated alert mechanisms.

Among these features, the SOS alert system is especially important, as it ensures immediate communication during emergencies. Designing a system focused on real-time threat detection and automatic SOS activation improves reliability and response efficiency in safety applications. However, many existing women safety systems rely on complex hardware, consume high power, or lack intelligent decision-making capabilities. Some systems only record or track events without actively detecting threats, which limits their effectiveness during critical situations.

To address these challenges, the proposed Ecliptic Watchdog system introduces a compact IoT-based surveillance solution built around a Raspberry Pi. It integrates a camera module, motor drivers, and DC motors for dynamic monitoring and wider coverage. Using Python-based intelligent algorithms, the system analyzes real-time data and automatically triggers an SOS alert when a threat is detected. This active threat-detection approach ensures faster response, structural simplicity, and improved adaptability across different environments. The rest of this report is organized as follows: Section II presents the system architecture, Section III explains the implementation, Section IV discusses the alert and motor control integration, Section V provides performance analysis, and Section VI concludes the project with future scope.

II. METHOD

The proposed system follows a structured approach for designing and implementing an IoT-based women safety surveillance solution. The system is built around a Raspberry Pi, which acts as the central

controller for data processing and decision-making.

Figure 1 shows the hardware arrangement, including the Raspberry Pi, camera module, motor drivers, and DC motors. The camera captures real-time video, while the motors enable wider area coverage through controlled movement. The captured data is processed using Python-based intelligent algorithms to detect suspicious activities. If a threat is identified, the system immediately triggers an SOS alert for quick response.

Figure 2 presents the system architecture, which consists of three main stages: data acquisition, processing and decision-making, and emergency response. This method ensures real-time monitoring, fast detection, and reliable safety performance.

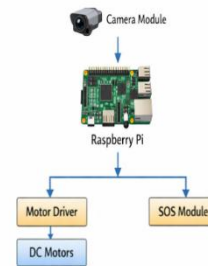


Fig. 1: Hardware setup of the proposed system.

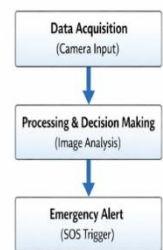


Fig. 2: Block diagram of the system architecture.

III. Construction of Smart IoT Surveillance Structure

The Smart IoT Surveillance System is a key part of the proposed framework for protecting women. It aims to detect emergencies and immediately notify trusted contacts or authorities. The system functions as an intelligent monitoring unit. It combines embedded hardware with Internet of Things (IoT) communication technologies. Its main goal is to provide real-time

safety support through location tracking, alert messaging, and audible warnings.

In the proposed design, the surveillance structure uses a microcontroller-based embedded platform. This platform coordinates the operation of sensing, processing, and communication modules. The system includes various hardware components like a GPS module, GSM communication module, panic button, buzzer, and power management unit. These components work together to detect emergencies and send alert signals over wireless networks.

The microcontroller serves as the central control unit. It receives input from the panic button and manages communication between the GPS and GSM modules. When the user presses the emergency button, the microcontroller activates the GPS module to get the user's real-time location. The GPS module finds the latitude and longitude using satellite technology. It then processes these coordinates and sends them to the GSM communication module.

The GSM module sends alert messages to predefined contacts, such as family members, friends, or emergency authorities. The alert message usually includes a warning and the user's exact location. This helps recipients quickly find the user's position and offer immediate help. The communication process is designed to work efficiently and with minimal delay, ensuring alerts are delivered promptly.

The system also includes a buzzer alarm to provide local alerts during emergencies. When the panic button is pressed, the buzzer emits a loud

signal to attract nearby attention. This feature increases the chances of receiving immediate help from people nearby. The system's overall architecture is shown in

Fig. 3 (Block Diagram of the Smart IoT Surveillance System).

The block diagram illustrates how the input module, processing unit, location tracking module, and communication module interact. The panic button is the main input device, while the microcontroller processes the signal and coordinates the activation of other modules. To evaluate how well the proposed system works, the hardware model is built and tested using embedded development tools and communication modules.

The microcontroller firmware is programmed to continuously monitor input signals and activate the emergency protocol as needed. The communication module is set up to send SMS alerts and location data to registered contacts. Testing the system is crucial for assessing the reliability and performance of the proposed surveillance device.

Different operational scenarios are tested to check response time, communication accuracy, and system stability. Performance metrics such as alert transmission delay, GPS accuracy, and power consumption are analyzed to ensure efficient operation.

Table I lists the key hardware parameters used in the Smart IoT Surveillance System. These parameters outline the operational conditions for the embedded modules and communication components.

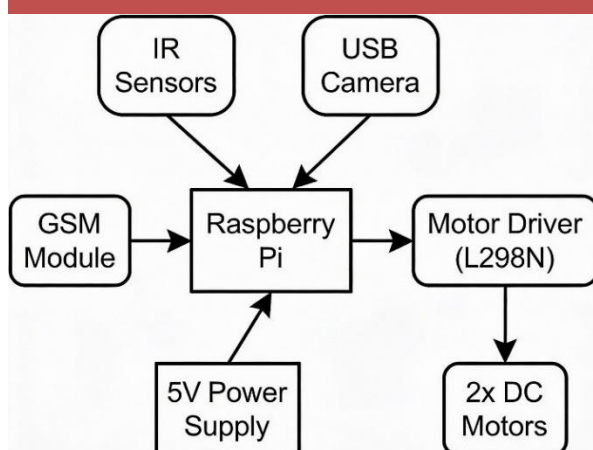


Fig 3: Block diagram of Smart IoT Surveillance

Parameters	System Configuration
Microcontroller	Embedded Control Unit
GPS Module	Real-Time Location Tracking
GSM Module	SMS Alert Communication
Panic Button	Emergency Trigger
Buzzer	Audible Alert
Operating Voltage	5V
Communication Type	GSM Network

In summary, the proposed IoT-based Night Patrolling System provides significant advantages over traditional surveillance methods. It offers real-time monitoring, automated detection of unauthorized individuals, and rapid alert generation, ensuring improved response time and enhanced security. The system is energy-efficient, cost-effective, and easily deployable in various environments such as campuses, residential areas, and public spaces. These features make it a reliable solution for continuous surveillance and an effective foundation for developing advanced smart security systems. Furthermore, it can be extended to applications such as smart city monitoring, automated threat detection, and intelligent

safety networks, contributing to safer and more secure environments.

IV. Design of IoT Based Surveillance System

The design of the IoT-based Patrolling System plays a crucial role in developing an efficient and intelligent surveillance solution for women's safety. The system is designed to monitor an area continuously, detect the presence of individuals, and identify whether they are authorized or unauthorized. It processes multiple inputs such as live video feed, facial data, and predefined datasets, and produces outputs in the form of detection results and alert notifications.

Building upon the core concepts of IoT and computer vision, the proposed system integrates a Raspberry Pi-based processing unit with a webcam and face recognition algorithms to achieve real-time monitoring. The design emphasizes accurate detection, fast response, and reliable alert mechanisms. The entire system operates autonomously, reducing the need for manual intervention.

The effectiveness of the system relies on the ability of face recognition algorithms to identify individuals by comparing captured images with a pre-trained dataset. Using techniques such as Haar Cascade for face detection and LBPH (Local Binary Pattern Histogram) for recognition, the system ensures efficient processing and accurate classification. By properly integrating these components, the system achieves continuous surveillance and instant alert generation, making it a practical solution for enhancing security in real-world environments.

The alert generation process in the proposed system is defined based on the detection and recognition results obtained from the face recognition module. When a detected face does not match with the stored dataset of authorized individuals, the system classifies it as an unauthorized entry and triggers an alert. This decision mechanism can be represented as:

Alert = NOT (Authorized Match)

This logic ensures that alerts are generated only when an unknown individual is identified, thereby reducing false notifications. The implementation of this logic is achieved through the integration of image processing and decision-making modules. The system continuously analyses video frames captured from the webcam, and the presence or absence of a match determines the subsequent action. If a match is found, the system continues monitoring; otherwise, it activates the alert module.

The construction of the overall system begins with integrating individual functional modules such as face detection, face recognition, and alert generation. Separate processing flows are maintained for detection and notification to ensure efficient operation and minimal delay in response. Proper configuration of system parameters, including camera resolution, frame rate, dataset quality, and recognition thresholds, is essential for achieving accurate and reliable performance. These parameters directly influence detection accuracy, processing speed, and system stability.

The complete system is developed and tested using Python and OpenCV, which provide robust tools for real-time image processing. The modules are interconnected based on the designed

workflow, ensuring smooth data flow from image acquisition to alert generation. Both real-time and continuous monitoring scenarios are tested to validate the system's performance. The evaluation focuses on parameters such as detection accuracy, response time, and reliability of alert notifications. The system successfully identifies unauthorized individuals and generates alerts with minimal delay.

The results demonstrate that the system operates efficiently under normal conditions, producing accurate detection results and timely alerts. By optimizing system parameters and ensuring proper integration of modules, the proposed system achieves reliable performance suitable for real-world surveillance and security applications.

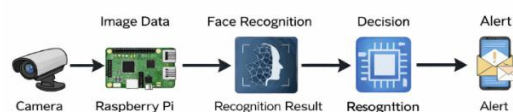


Fig 4: System Architecture of Patrolling System

Overall, the proposed IoT-based Patrolling System offers several advantages over traditional surveillance and monitoring methods. It operates using real-time data processing and automated decision-making, enabling faster detection and response to potential threats. The system is energy-efficient, cost-effective, and easily deployable across different environments. The integration of computer vision with IoT ensures scalability and flexibility, allowing the system to be expanded with additional sensors and advanced features. Due to its reliable performance and continuous monitoring capability, the proposed system serves as a strong foundation for developing intelligent security solutions, including smart surveillance

networks, automated threat detection systems, and advanced safety applications.

V. RESULT AND DISCUSSION

The results obtained from the design and implementation of the proposed IoT-based Patrolling System demonstrate that the system operates effectively and reliably for real-time surveillance applications. Extensive testing was conducted under different environmental conditions to evaluate both the functional accuracy and overall system performance.

The implementation was carried out using Python and OpenCV on a Raspberry Pi platform to ensure practical feasibility.

Various performance parameters were analyzed to validate the effectiveness of the system, including detection accuracy, response time, and reliability of alert generation. The experimental results confirm that the system accurately identifies unauthorized individuals and generates alerts with minimal delay. These observations indicate that the proposed design is suitable for real-world security applications and can be effectively deployed in intelligent surveillance systems.

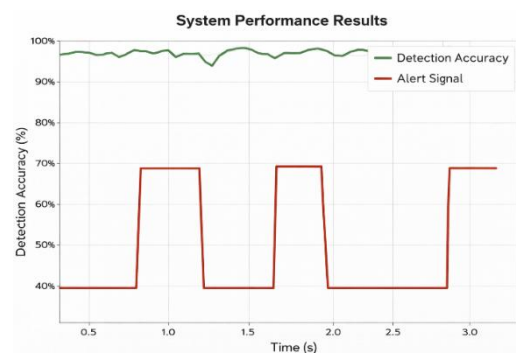


Fig 5: System Performance Results

The experimental results confirm that

the proposed IoT-based Patrolling System accurately detects and classifies individuals under various conditions. The system's behavior aligns with the expected operational logic, clearly distinguishing between authorized and unauthorized individuals. The integration of face detection and recognition algorithms ensures reliable performance and consistent output. Key performance parameters such as detection accuracy, response time, and alert reliability were evaluated to assess system efficiency and stability. The results demonstrate that the system maintains high accuracy and generates timely alerts, making it suitable for real-time surveillance and security applications.

VI. CONCLUSION

This work addressed the limitations of traditional surveillance systems and proposed an IoT-based Patrolling System that integrates real-time monitoring with face recognition technology. By leveraging computer vision techniques and IoT capabilities, the system enables automated detection of unauthorized individuals and instant alert generation, improving response time and overall security. The implementation using Raspberry Pi and OpenCV ensures a cost-effective and scalable solution suitable for real-world deployment.

The system was successfully designed and tested under various conditions, where experimental results confirmed accurate detection, reliable classification, and timely alert notifications. Performance evaluation demonstrated high detection accuracy, minimal response delay, and stable operation, validating the effectiveness

of the proposed approach for intelligent surveillance applications.

Despite these advantages, certain challenges such as performance in low-light conditions, dependency on dataset quality, and network requirements for alert transmission still exist. Future work can focus on integrating deep learning models for improved accuracy, incorporating mobile-based notifications, and enhancing system performance using cloud and edge computing technologies.

In conclusion, the proposed system provides an efficient, automated, and reliable solution for modern surveillance needs. It lays a strong foundation for developing advanced smart security systems, including intelligent monitoring networks and smart city safety applications, contributing to a safer and more secure environment.

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AUTHOR CONTRIBUTIONS STATEMENT

The authors confirm that they were fully responsible for all aspects of this research. They designed and developed the IoT-based Patrolling System, including the system architecture and implementation methodology. They carried out the development using Raspberry Pi, Python, and OpenCV, conducted experimental testing, and

analyzed the system performance. Additionally, the authors were responsible for interpreting the results, preparing the original manuscript, and reviewing and refining the final version of the paper.

CONFLICT OF INTEREST STATEMENT

The authors state that there are no known financial interests, personal relationships, or affiliations that could have affected the work presented in this paper. Additionally, the authors confirm that there are no political, religious, ideological, academic, or intellectual conflicts of interest related to this research.

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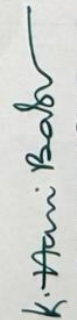


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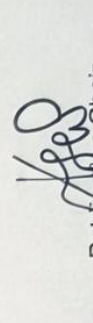
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**Ecliptic Watchdogs of Intelligent Perception Smart IoT
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